

5G RAN

Turbo Uplink (Low-Frequency TDD) Feature Parameter Description

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Huawei Technologies Co., Ltd.

Address: Huawei Industrial Base
Bantian, Longgang
Shenzhen 518129
People's Republic of China

Website: <https://www.huawei.com>

Email: support@huawei.com

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1 Change History

The following describes the specific changes in different issues.

5G RAN10.1 Draft A (2025-12-31)

This issue introduces the following changes to 5G RAN9.1 05 (2025-12-14).

- Technical changes

Change Description	Parameter Change	RAT	Base Station Model
Added support for QCI-specific uplink data compression. For details, see: • 5.1.4 Uplink Data Compression • 5.3.2.1 Prerequisite Functions • 5.3.2.2 Mutually Exclusive Functions • 5.3.2.3 Function Impacts • 5.4.1.1 Data Preparation • 5.4.1.2 Using MML Commands	Modified parameter: Added the QCI_UDC_SW option to the gNBQciBearer.PdcpAlgoSwitch parameter.	Low-frequency TDD	3900 and 5900 series base stations (macro base stations)
Added an impact relationship between closed-loop PUSCH antenna selection and uplink precise MCS optimization. For details, see 4.3.2.3 Function Impacts .	None	Low-frequency TDD	3900 and 5900 series base stations (macro base stations)

Change Description	Parameter Change	RAT	Base Station Model
Added an impact relationship between probability-based uplink MCS selection and uplink precise MCS optimization. For details, see 5.3.2.3 Function Impacts .	None	Low-frequency TDD	3900 and 5900 series base stations (macro base stations)
Added an impact relationship between uplink 4-layer transmission for a single FWA user and high-reliability uplink MCS selection. For details, see 5.3.2.3 Function Impacts .	None	Low-frequency TDD	3900 and 5900 series base stations (macro base stations)
Added an impact relationship between uplink 4-layer transmission for a single FWA user and full-power dual transmission. For details, see 4.3.2.3 Function Impacts .	None	Low-frequency TDD	3900 and 5900 series base stations (macro base stations)
Added an impact relationship between uplink 4-layer transmission for a single FWA user and spatial channel estimation under weak uplink coverage. For details, see 4.3.2.3 Function Impacts .	None	Low-frequency TDD	3900 and 5900 series base stations (macro base stations)
Added an impact relationship between uplink 4-layer transmission for a single FWA user and closed-loop PUSCH antenna selection. For details, see 4.3.2.3 Function Impacts .	None	Low-frequency TDD	3900 and 5900 series base stations (macro base stations)

- Editorial changes

Revised descriptions in this document, which do not involve changes in meanings and therefore are not detailed.

2 About This Document

2.1 General Statements

Purpose

Feature Parameter Description documents are intended to acquaint readers with:

- The technical principles of features and their related parameters
- The scenarios where these features are used, the benefits they provide, and the impact they have on networks and functions
- Requirements that must be met before feature activation
- Parameter configuration required for feature activation, verification of feature activation, and monitoring of feature performance

NOTE

This document only provides guidance for feature activation. Feature deployment and feature gains depend on the specifics of the network scenario where the feature is deployed. To achieve optimal gains, contact Huawei professional service engineers.

Functions mentioned in this document work properly only when enabled in the specified applicable scenarios (such as RAT and networking). If a function not mentioned in this document is enabled or a function is enabled in a scenario not specified as applicable, exceptions or other impacts may occur.

Software Interfaces

Any parameters, alarms, counters, or managed objects (MOs) described in Feature Parameter Description documents apply only to the corresponding software release. For future software releases, refer to the corresponding updated product documentation.

2.2 Features in This Document

This document describes the following features.

Feature ID	Feature Name	Chapter/Section
FOFD-06120 5	Turbo Uplink	4 Turbo Uplink
FOFD-08120 2	Turbo Uplink 2.0	5 Turbo Uplink 2.0

2.3 Differences

Differences Between NSA and SA

Function Name	Difference	Chapter/Section
RRC-reconfiguration-based dynamic waveform switching	None	4.1.1 RRC-Reconfiguration-based Dynamic Waveform Switching
Frequency selective scheduling under weak uplink coverage	None	4.1.2 Frequency Selective Scheduling Under Weak Uplink Coverage
Spatial channel estimation under weak uplink coverage	None	4.1.3 Spatial Channel Estimation Under Weak Uplink Coverage
SR-based scheduling optimization	None	4.1.4 SR-based Scheduling Optimization
Uplink SINR filtering optimization	None	4.1.5 Uplink SINR Filtering Optimization
Full-power dual transmission	Supported only in SA networking	4.1.6 Full-Power Dual Transmission
Uplink $\pi/2$ -BPSK modulation	None	4.1.7 Uplink $\pi/2$-BPSK Modulation
Uplink $\pi/2$ -BPSK modulation power boost	Supported only in SA networking	4.1.8 Uplink $\pi/2$-BPSK Modulation Power Boost
Uplink low-PAPR DMRS	Supported only in SA networking	4.1.9 Uplink Low-PAPR DMRS
Closed-loop PUSCH antenna selection	Supported only in SA networking	4.1.10 Closed-Loop PUSCH Antenna Selection

Function Name	Difference	Chapter/Section
Interference randomization under weak uplink coverage	None	5.1.1 Interference Randomization Under Weak Uplink Coverage
DMRS interference reduction	None	5.1.2 DMRS Interference Reduction
High-reliability uplink MCS selection	None	5.1.3 High-Reliability Uplink MCS Selection
Uplink data compression	Supported only in SA networking	5.1.4 Uplink Data Compression
Probability-based uplink MCS selection	None	5.1.5 Probability-based Uplink MCS Selection
Time-domain adaptive filtering	None	5.1.6 Time-Domain Adaptive Filtering

Differences Between High Frequency Bands and Low Frequency Bands

This document refers to frequency bands belonging to FR1 (410–7125 MHz) as low frequency bands, and those belonging to FR2 (24250–71000 MHz) as high frequency bands. For details about FR1 and FR2, see section 5.1 "General" in 3GPP TS 38.104 V17.8.0.

Function Name	Difference	Chapter/Section
RRC-reconfiguration-based dynamic waveform switching	Supported in both high and low frequency bands. The differences are as follows: Different thresholds are used for switching between the CP-OFDM and DFT-s-OFDM waveforms. RRC-reconfiguration-based dynamic waveform switching in low frequency bands is described in this document while that in high frequency bands is described in <i>Ultra Uplink (High-Frequency TDD)</i> .	4.1.1 RRC-Reconfiguration-based Dynamic Waveform Switching

Function Name	Difference	Chapter/Section
Frequency selective scheduling under weak uplink coverage	Supported only in low frequency bands	4.1.2 Frequency Selective Scheduling Under Weak Uplink Coverage
Spatial channel estimation under weak uplink coverage	Supported only in low frequency bands	4.1.3 Spatial Channel Estimation Under Weak Uplink Coverage
SR-based scheduling optimization	Supported in both high and low frequency bands. SR-based scheduling optimization in low frequency bands is described in this document while that in high frequency bands is described in <i>Uplink Scheduling</i> .	4.1.4 SR-based Scheduling Optimization
Uplink SINR filtering optimization	Supported only in low frequency bands	4.1.5 Uplink SINR Filtering Optimization
Full-power dual transmission	Supported in both high and low frequency bands. This document describes only full-power dual transmission in low-frequency NR TDD. For details about this function in high-frequency NR TDD, see the section "Full-Power Dual Transmission" in <i>Ultra Uplink (High-Frequency TDD)</i> .	4.1.6 Full-Power Dual Transmission
Uplink $\pi/2$ -BPSK modulation	Supported only in low frequency bands	4.1.7 Uplink $\pi/2$-BPSK Modulation
Uplink $\pi/2$ -BPSK modulation power boost	Supported only in low frequency bands	4.1.8 Uplink $\pi/2$-BPSK Modulation Power Boost
Uplink low-PAPR DMRS	Supported only in low frequency bands	4.1.9 Uplink Low-PAPR DMRS
Closed-loop PUSCH antenna selection	Supported only in low frequency bands	4.1.10 Closed-Loop PUSCH Antenna Selection

Function Name	Difference	Chapter/Section
Interference randomization under weak uplink coverage	Supported only in low frequency bands	5.1.1 Interference Randomization Under Weak Uplink Coverage
DMRS interference reduction	Supported only in low frequency bands	5.1.2 DMRS Interference Reduction
High-reliability uplink MCS selection	Supported only in low frequency bands	5.1.3 High-Reliability Uplink MCS Selection
Uplink data compression	Supported in both high and low frequency bands. This document describes only uplink data compression in low-frequency NR TDD. For details about this function in high-frequency NR TDD, see the section "Uplink Data Compression" in <i>Ultra Uplink (High-Frequency TDD)</i> .	5.1.4 Uplink Data Compression
Probability-based uplink MCS selection	Supported only in low frequency bands	5.1.5 Probability-based Uplink MCS Selection
Time-domain adaptive filtering	Supported only in low frequency bands	5.1.6 Time-Domain Adaptive Filtering

3 Overview

Uplink coverage is affected by factors such as propagation loss, wall penetration loss, and interference. The higher the NR frequency, the greater the impact on coverage and the higher the probability that weak coverage occurs, especially in indoor areas. In addition, uplink coverage is affected by the limited maximum transmit power of UEs. **To enhance uplink coverage, Turbo Uplink is introduced. It consists of turbo power and turbo receiver, with the former used to increase the transmit power of UEs and the latter used to improve the demodulation capability of base station receivers. In this way, it increases the user-perceived rate and improves user experience in weak coverage areas.** The following describes the features currently included in the Turbo Uplink solution.

- Turbo Uplink: This feature is controlled by the **TURBO_UPLINK_SW** option of the **NRDUCellChnCovAlgo.UlCoverageAlgoSwitch** parameter. Functions of the Turbo Uplink feature (which are listed in [Table 3-1](#)) can take effect only after this option is selected. For MML commands for enabling and disabling the Turbo Uplink feature, see [4.4.1.2 Using MML Commands](#). To maximize the gains provided by this feature, related parameters need to be configured before feature activation. For details about the configuration, see [4.4.1.1 Data Preparation](#) and [4.4.1.2 Using MML Commands](#).

Table 3-1 Functions of the Turbo Uplink feature

Catego ry	Function Name	Principle Overview	Reference
Turbo power	RRC- reconfigurat ion-based dynamic waveform switching	This function enables a base station to adaptively instruct UEs to switch from the CP-OFDM waveform to the DFT-s-OFDM waveform as the maximum power reduction (MPR) allowed with the DFT-s-OFDM waveform is lower than that allowed with the CP-OFDM waveform. This reduces the amount of power reduction and increases the actual transmit power of UEs.	4.1.1 RRC-Reconfiguration-based Dynamic Waveform Switching

Category	Function Name	Principle Overview	Reference
	Full-power dual transmission	This function enables 2T UEs to transmit data at full power when the UEs are working in rank 1 at the cell edge.	4.1.6 Full-Power Dual Transmission
	Uplink $\pi/2$ -BPSK modulation	These functions reduce the peak to average power ratio (PAPR) of PUSCH and DMRS and increase the transmit power of UEs by introducing $\pi/2$ -BPSK as the PAPR of $\pi/2$ -BPSK is lower than that of other modulation schemes.	4.1.7 Uplink $\pi/2$-BPSK Modulation
	Uplink $\pi/2$ -BPSK modulation power boost		4.1.8 Uplink $\pi/2$-BPSK Modulation Power Boost
	Uplink low-PAPR DMRS		4.1.9 Uplink Low-PAPR DMRS
Turbo Receiver	Frequency selective scheduling under weak uplink coverage	This function schedules UEs in RB positions with high SINRs based on the frequency selective characteristics of channels and increases the transport block size for scheduling to increase the uplink throughput of UEs.	4.1.2 Frequency Selective Scheduling Under Weak Uplink Coverage
	Spatial channel estimation under weak uplink coverage	This function uses multiple antennas to identify spatial characteristics of wanted signals and interference signals. Based on the identified spatial characteristics, this function adaptively adjusts receive beams of the base station to reduce the interference to target UEs along the uplink interference direction.	4.1.3 Spatial Channel Estimation Under Weak Uplink Coverage
	SR-based scheduling optimization	This function preferentially schedules UEs that report a non-zero buffer size, ensuring that UEs with data scheduling requirements are scheduled as soon as possible.	4.1.4 SR-based Scheduling Optimization
	Uplink SINR filtering optimization	This function selects large filtering coefficients for uplink SINRs to match uplink channel changes when uplink channels fluctuate greatly.	4.1.5 Uplink SINR Filtering Optimization

Category	Function Name	Principle Overview	Reference
	Closed-loop PUSCH antenna selection	This function enables the base station to instruct UEs to select the optimal antenna for uplink data transmission based on SRS measurement results, increasing the uplink throughput of UEs.	4.1.10 Closed-Loop PUSCH Antenna Selection

- Turbo Uplink 2.0: As the NR traffic volume increases, inter-cell interference also increases, affecting user experience. To further improve user experience under weak coverage, the Turbo Uplink 2.0 feature is introduced. This feature is controlled by the **TURBO_UPLINK_PHASE2_SW** option of the **NRDUCellChnCovAlgo.UlCoverageAlgoSwitch** parameter. Functions of the Turbo Uplink 2.0 feature (which are listed in [Table 3-2](#)) can take effect only after this option is selected. For MML commands for enabling and disabling the Turbo Uplink 2.0 feature, see [5.4.1.2 Using MML Commands](#). It is recommended that the functions involved in Turbo Uplink 2.0 be enabled together. To maximize the gains provided by this feature, related parameters need to be configured before feature activation. For details about the configuration, see [5.4.1.1 Data Preparation](#) and [5.4.1.2 Using MML Commands](#).

Table 3-2 Functions of the Turbo Uplink 2.0 feature

Category	Function Name	Principle Overview	Reference
Turbo receiver	Interference randomization under weak uplink coverage	This function schedules UEs under weak coverage in RB positions with high SINRs based on the frequency selective characteristics of channels. In addition, this function staggers the start positions of RBs for cells with the same result of PCI mod 3 to reduce inter-cell uplink interference.	5.1.1 Interference Randomization Under Weak Uplink Coverage
	DMRS interference reduction	This function adaptively distinguishes UEs with strong interference from UEs with weak interference and allocates different DMRS resources to the two types of UEs to reduce DMRS resource collisions between UEs.	5.1.2 DMRS Interference Reduction

Category	Function Name	Principle Overview	Reference
	High-reliability uplink MCS selection	This function uses both SRS SINR and DMRS SINR measurement results to improve the reliability of uplink MCS selection and increase the uplink user-perceived rate.	5.1.3 High-Reliability Uplink MCS Selection
	Uplink data compression	This function uses the 3GPP Deflate compression algorithm to compress duplicate contents in uplink data packets. This reduces resource consumption over the air interface and improves uplink data transmission capability.	5.1.4 Uplink Data Compression
	Probability-based uplink MCS selection	The base station estimates the channel quality fluctuation level based on the probability of different SINRs at historical measurement and scheduling moments. This makes uplink MCS selection more reliable for UEs and increases their uplink user-perceived rates.	5.1.5 Probability-based Uplink MCS Selection
	Time-domain adaptive filtering	The base station adaptively adjusts time-domain filtering coefficients based on the power delay profile (PDP) of PUSCH DMRSs and the time-domain changes in the strength of useful signals and interference signals on each RX channel. This increases the SINR of received signals.	5.1.6 Time-Domain Adaptive Filtering

4 Turbo Uplink

4.1 Principles

4.1.1 RRC-Reconfiguration-based Dynamic Waveform Switching

NR supports two waveforms: Cyclic Prefix-OFDM (CP-OFDM) and DFT-spread OFDM (DFT-s-OFDM).

- CP-OFDM allows for the use of non-contiguous frequency-domain resources and flexible resource allocation. It brings a high frequency-diversity gain but produces a high PAPR.
- DFT-s-OFDM features a low PAPR (close to that of a single carrier) and high transmit power. However, it allows for the use of only contiguous frequency-domain resources.

The RRC-reconfiguration-based dynamic waveform switching function enables a base station to adaptively instruct UEs to switch from the CP-OFDM waveform to the DFT-s-OFDM waveform, reducing the PAPR and increasing the transmit power of UEs. This function is controlled by the **UL_WAVEFORM_ADAPT_SW** option of the **NRDUCellAlgoSwitch.AdaptiveEdgeExpEnhSwitch** parameter.

Before this function is enabled, the network side instructs UEs to always select CP-OFDM.

After this function is enabled, the network side adaptively instructs UEs to select CP-OFDM or DFT-s-OFDM waveform based on the radio environment and the threshold specified by the **NRDUCellPusch.SinrThldforWaveformSel** parameter. In low-frequency scenarios, if the uplink SINR of a UE is less than the threshold and rank 1 is used, the UE selects DFT-s-OFDM; otherwise, the UE selects CP-OFDM. One waveform can be switched to the other waveform through the *transformPrecoder* IE in the RRCReconfiguration message.

After this function is enabled, the service drop rate may increase if there are lost UEs in a cell, because these UEs cannot respond to the RRC reconfiguration messages for uplink waveform switching. (Lost UEs refer to the UEs that

voluntarily stop sending data on the NR PUSCH but have not been released due to timeout by the core network.) In this case, it is recommended that the function of lost-UE protection against service drops be enabled by selecting the **LOST_UE_DROP_PROTECT_SW** option of the **NRDUCellUIMc.UlAmcPerformanceSw** parameter. With this option selected, the base station optimizes the uplink waveform switching procedure for the lost UEs and reduces the number of RRC reconfiguration messages for uplink waveform switching. This decreases the service drop rate in the cell.

 NOTE

To prevent ping-pong waveform switching after RRC-reconfiguration-based dynamic waveform switching is enabled, this function automatically sets a hysteresis for the threshold.

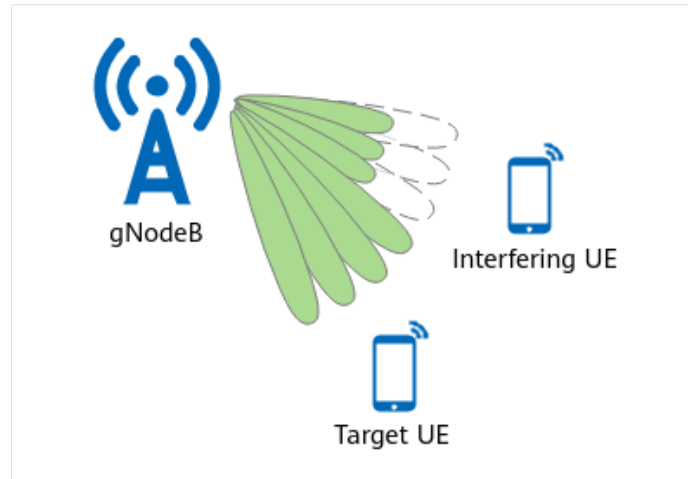
4.1.2 Frequency Selective Scheduling Under Weak Uplink Coverage

Frequency selective scheduling under weak uplink coverage is introduced to address the issue that the user-perceived rate may be affected in weak coverage areas as the transmit power of UEs is limited. This function is controlled by the **UL_WEAK_COV_FREQ_SEL_SCH_SW** option of the **NRDUCellChnCovAlgo.UlCoverageAlgoSwitch** parameter. After this function is enabled, it utilizes the frequency selective characteristics of channels (that is, different SINRs in different frequency-domain positions). It schedules UEs in RB positions with high SINRs and increases the transport block size. In this way, the function increases the average uplink UE throughput in weak coverage areas. This function can take effect for a UE only when the UE is under weak coverage, the power is limited, and the buffered data volume is large enough.

4.1.3 Spatial Channel Estimation Under Weak Uplink Coverage

Spatial channel estimation under weak uplink coverage uses multiple antennas to identify spatial characteristics of signals from target UEs and interference signals from neighboring cells. Based on the identified spatial characteristics, this function adaptively adjusts receive beams of the base station to reduce the interference to target UEs along the uplink interference direction, as shown in [Figure 4-1](#). This improves the accuracy of channel estimation, and increases the uplink SINR and average uplink throughput for UEs in weak coverage areas. This function is controlled by the **UL_WEAK_COV_SPACE_CHN_ESTI_SW** option of the **NRDUCellChnCovAlgo.UlCoverageAlgoSwitch** parameter. This function takes effect for UEs with uplink data transmission only when the UEs are under weak coverage, the modulation scheme is QPSK or $\pi/2$ -BPSK, the transmission rank is 1, and the type of DMRS is type 1.

Figure 4-1 Principles of spatial channel estimation under weak uplink coverage



4.1.4 SR-based Scheduling Optimization

SR-based scheduling optimization is introduced to promptly schedule UEs that have data to be scheduled when UEs with scheduling requests (SRs) occupy a large number of air interface resources in uplink heavy load scenarios. This function is controlled by the **SR_SCH_OPT_SW** option of the **NRDUCellPusch.PuschPerformanceSwitch** parameter. After SR-based scheduling optimization is enabled, the gNodeB lowers the scheduling priorities of UEs with zero buffer size and preferentially schedules UEs with non-zero buffer size. To schedule as much uplink data of a UE with non-zero buffer size as possible at a time, the gNodeB also adjusts the first packet volume based on the historical packet transmission status of the UE.

4.1.5 Uplink SINR Filtering Optimization

The SINR of an uplink channel reflects the quality of this channel. If the SINR at the gNodeB measurement moment differs greatly from that at the scheduling moment, the uplink BLER will increase, affecting uplink user experience. To address this issue, uplink SINR filtering optimization is introduced for moving UEs that require a long time to complete SRS frequency hopping over the entire bandwidth. **This function uses large filtering coefficients for uplink SINRs so that the uplink SINRs filtered are greatly determined by those at the current measurement moment but slightly determined by those at historical measurement moments.** This adapts to fast changes of uplink channels and increases the uplink user-perceived rate. This function is controlled by the **UL_SINR_FILTER_OPT_SW** option of the **NRDUCellUIMc.UIMcAlgoSw** parameter.

4.1.6 Full-Power Dual Transmission

Full-power dual transmission is introduced to improve the performance of cell-edge UEs in SA networking. **This function enables UEs with two or more transmit antennas and SRS ports to transmit at full power when the UEs work in rank 1,** achieving better performance in SA networking. This function is controlled by the **UL_RANK1_FULL_POWER_SW** option of the **NRDUCellAlgoSwitch.AdaptiveEdgeExpEnhSwitch** parameter. After full-power dual transmission is enabled, it works as follows:

1. The base station checks the *ul-FullPwrMode-r16*, *ul-FullPwrMode1-r16*, and *ul-FullPwrMode2-TPMIGroup-r16* fields in the *FeatureSetUplink* IE to determine whether a UE supports full-power dual transmission. For details, see [4.3.5 Others](#).
2. If the UE supports this function, the base station sets *ul-FullPowerTransmission-r16* in *PUSCH-Config* to *fullpower*, *fullpowerMode1*, or *fullpowerMode2* for full-power dual transmission to take effect for the UE. If the UE reports two or more capability fields at the same time, the base station instructs the UE to use the transmission modes in the following priorities: *fullpower* > *fullpowerMode2* > *fullpowerMode1*.
3. After this configuration, full-power dual transmission takes effect.

 NOTE

After full-power dual transmission is enabled, for UEs that support coherent codebooks and only the *fullpowerMode1* transmission mode (as described in [4.3.5 Others](#)), coherent codebooks instead of full-power dual transmission take effect. The coherent codebook capability is indicated by the value **fullCoherent** of *pusch-TransCoherence* in the *MIMO-ParametersPerBand* IE reported by the UEs.

4.1.7 Uplink $\pi/2$ -BPSK Modulation

Uplink $\pi/2$ -BPSK modulation is introduced to improve the performance of edge UEs. **This function enables UEs to transmit at a higher power as $\pi/2$ -BPSK delivers a lower PAPR and allows lower power reduction (as defined in protocols) than other modulation schemes.** This function is controlled by the **UL_HALF_PI_BPSK_SW** option of the **NRDUCellAlgoSwitch.AdaptiveEdgeExpEnhSwitch** parameter. After uplink $\pi/2$ -BPSK modulation is enabled, it works as follows:

1. The base station determines whether a UE supports uplink $\pi/2$ -BPSK modulation based on the *pusch-HalfPi-BPSK* field in the *Phy-Parameters* IE.
2. The base station sets the *tp-pi2BPSK* field in the *pusch-config* IE to "enable" if the UE supports this function.
3. Uplink $\pi/2$ -BPSK modulation takes effect for the UE when the UE works with the DFT-s-OFDM waveform (which is described in [4.1.1 RRC-Reconfiguration-based Dynamic Waveform Switching](#)) and the MCS index is 0 or 1.

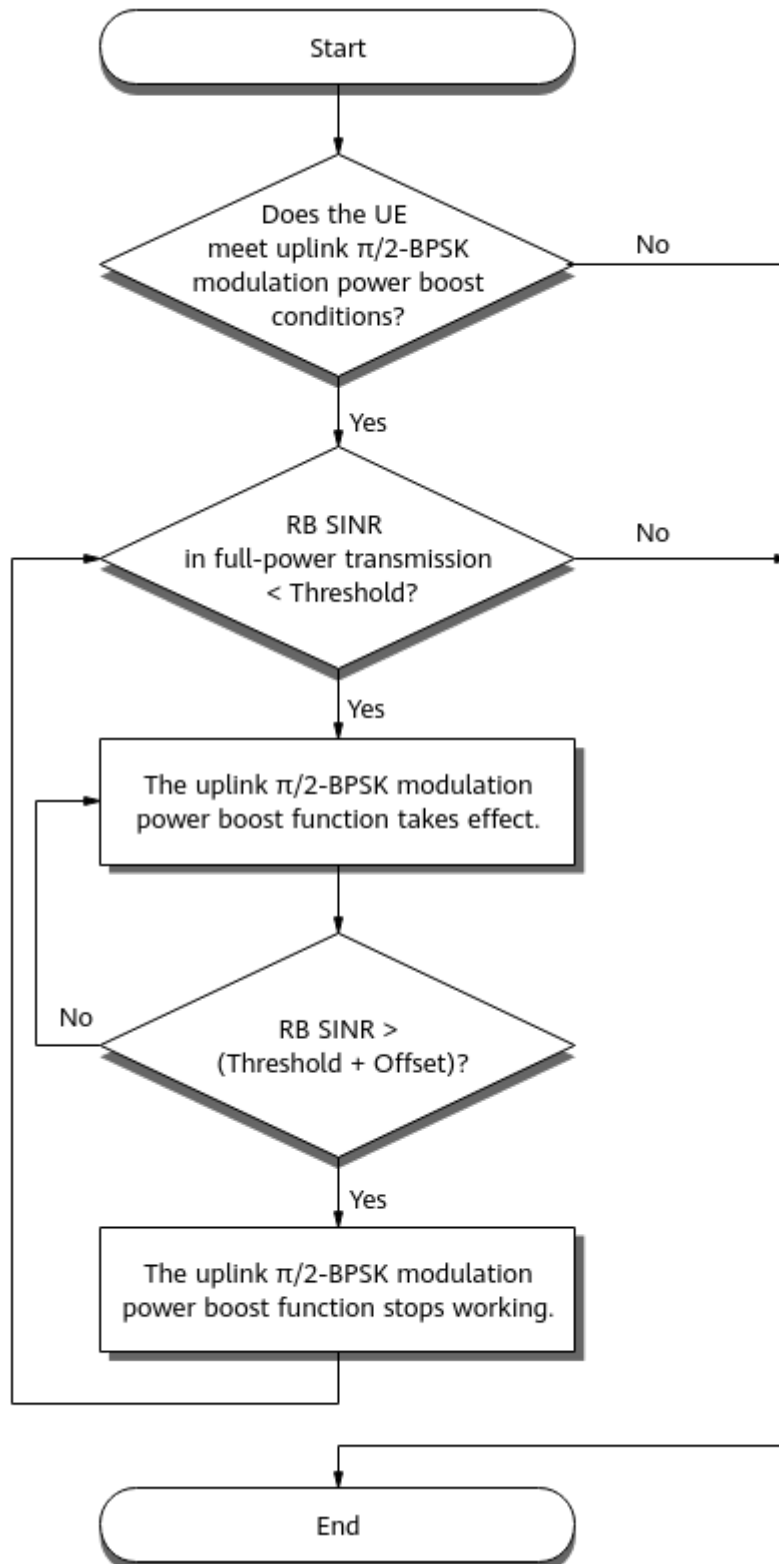
 NOTE

- For more information about this function, see the following specifications:
 - For details about MPR, see Table 6.2.2-1 "Maximum power reduction (MPR) for power class 3" and Table 6.2.2-2 "Maximum power reduction (MPR) for power class 2" in section 6.2.2 "UE maximum output power reduction" in 3GPP TS 38.101-1 V17.2.0.
 - For the MCS index tables, see Table 6.1.4.1-1 "MCS index table for PUSCH with transform precoding and 64QAM" and Table 6.1.4.1-2 "MCS index table 2 for PUSCH with transform precoding and 64QAM" in section 6.1.4 "Modulation order, redundancy version and transport block size determination" in 3GPP TS 38.214 V16.6.0.
- In cases where the **MSG3_TRANSFORMPRECODE_SWITCH** option of the **NRDUCellPusch.UlPuschAlgoSwitch** parameter is selected, you are not advised to enable uplink $\pi/2$ -BPSK modulation. Otherwise, if a UE performs contention-based access (in scenarios such as handover to the LTE side in NSA networking and resynchronization) after it initially accesses the network, indicators related to contention-based access and service drops deteriorate. A relevant analysis is provided as follows:
 - When only Msg3 transform precoding configuration is enabled, the waveform used for uplink transmission of Msg3 changes from CP-OFDM to DFT-s-OFDM. In this case, both the UE and the base station use the QPSK modulation scheme by default.
 - When only uplink $\pi/2$ -BPSK modulation function is enabled, the CP-OFDM waveform is used for uplink transmission of Msg3. In this case, both the UE and the base station use the QPSK modulation scheme.
 - When both uplink $\pi/2$ -BPSK modulation and Msg3 transform precoding configuration are enabled, the DFT-s-OFDM waveform is used for uplink transmission of Msg3. In this case, the UE may use the uplink $\pi/2$ -BPSK modulation scheme to transmit Msg3 for contention-based access (in scenarios such as handover to the LTE side in NSA networking and resynchronization) after it initially accesses the network, as the modulation scheme for Msg3 transmission in such contention-based access is not defined in the specifications. As the base station does not have the UE context, it uses the QPSK modulation scheme, which is different from the modulation scheme used by the UE. As a result, indicators related to contention-based access and service drops deteriorate.

4.1.8 Uplink $\pi/2$ -BPSK Modulation Power Boost

Uplink $\pi/2$ -BPSK modulation power boost is introduced to improve the performance of edge UEs. This function is controlled by the **UL_HALF_PI_BPSK_POWER_BOOST_SW** option of the **NRDUCellAlgoSwitch.AdaptiveEdgeExpEnhSwitch** parameter. [Figure 4-2](#) shows how the uplink $\pi/2$ -BPSK modulation power boost function takes effect for a UE after being enabled.

Figure 4-2 Triggering the uplink $\pi/2$ -BPSK modulation power boost function



A UE must meet the following conditions for uplink $\pi/2$ -BPSK modulation power boost to take effect.

- The UE supports $\pi/2$ -BPSK modulation and $\pi/2$ -BPSK modulation power boost. These capabilities are reported through corresponding fields.
 - $\pi/2$ -BPSK modulation: The UE reports its capability through the *pusch-HalfPi-BPSK* field in the *Phy-Parameters* IE.
 - $\pi/2$ -BPSK modulation power boost: The UE reports its capability through the *powerBoosting-pi2BPSK* field in the *RF-Parameters* IE.
- The UE supports power class 3 in one or more of TDD frequency bands n40/n41/n77/n78/n79.
- The UE uses no more than 40% of uplink symbols within one or more radio frames. (This proportion condition can be controlled through downlink-to-uplink slot configuration. For example, 7:3, 8:2, and 4:1 all meet this condition.)
- The maximum transmit power P_{EMAX} of the UE is greater than or equal to 20 dBm. For details about the maximum transmit power, see section 6.2.4 "Configured transmitted power" in 3GPP TS 38.101-1 V17.0.0.

Assume that the UE meets the preceding four conditions. When the base station estimates that the RB SINR will be less than the threshold specified by **NRDUCellUIMc.PwrBoostRbLimitSinrThld** if the UE performs transmission at full power, the uplink $\pi/2$ -BPSK modulation power boost function takes effect, **increasing the transmit power of the UE by 3 dB as $\pi/2$ -BPSK delivers a lower PAPR than other modulation schemes**. In this way, the function increases the uplink user-perceived rate when the UE is at the coverage edge. When the base station estimates that the RB SINR will be greater than the threshold plus the offset specified by **NRDUCellUIMc.PwrBoostRbLimitSinrThldOfs**, the uplink $\pi/2$ -BPSK modulation power boost function stops working. When the base station estimates that the RB SINR will be less than the threshold again, the function takes effect again.

Different UEs support different power boost values. If the value supported by a UE is not 3 dB, the base station can adjust the power boost value through the **NRDUCellUIMc.HalfPiBpskPwrBoostValue** parameter for consistency with the actual UE capability.

4.1.9 Uplink Low-PAPR DMRS

The uplink low-PAPR demodulation reference signal (DMRS) function is introduced to improve the performance of edge UEs. After this function is enabled, UEs work with the DFT-s-OFDM waveform (which is described in [4.1.1 RRC-Reconfiguration-based Dynamic Waveform Switching](#)) and **uplink low-PAPR DMRS sequences are used to reduce the DMRS PAPR and increase the DMRS transmit power**, thereby improving the uplink user-perceived rate of edge UEs. This function is controlled by the **UL_LOW_PAPR_DMRS_SW** option of the **NRDUCellAlgoSwitch.AdaptiveEdgeExpEnhSwitch** parameter.

After the uplink low-PAPR DMRS function is enabled, it works as follows:

1. The base station checks the *lowPAPR-DMRS-PUSCHwithPrecoding-r16* field in the *MIMO-ParametersPerBand* IE to determine whether a UE supports the uplink low-PAPR DMRS function.
2. If the UE supports this function, the base station sets the *dmrs-UplinkTransformPrecoding-r16* IE in *transformPrecodingEnabled* in *DMRS-UplinkConfig*.

3. After the UE completes the configuration, the uplink low-PAPR DMRS function takes effect to increase the DMRS transmit power when the UE works with the DFT-s-OFDM waveform and services are scheduled in $\pi/2$ -BPSK modulation mode.

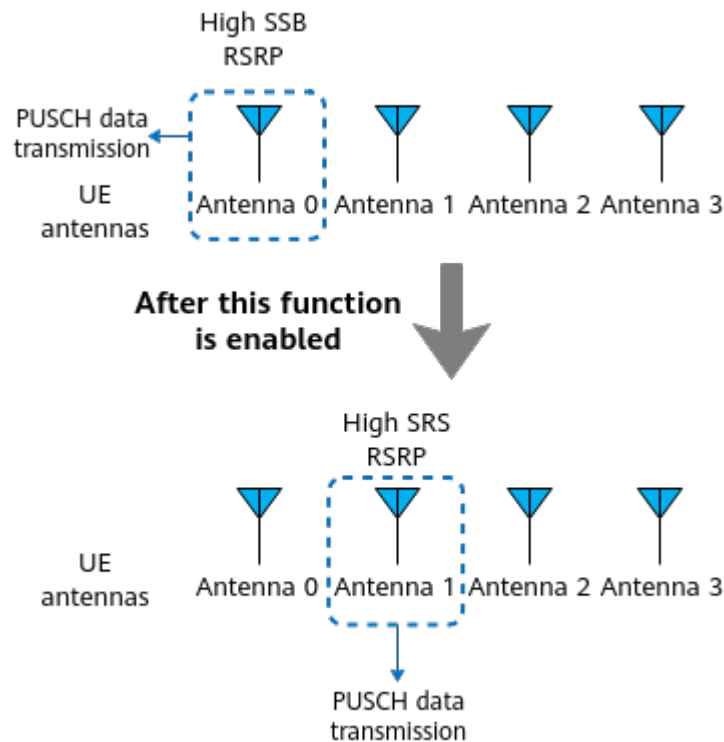
4.1.10 Closed-Loop PUSCH Antenna Selection

Conventional uplink antenna selection adopts an open-loop mechanism. Specifically, a UE selects uplink antennas based on the downlink SSB signal quality, as TDD systems have reciprocity between uplink and downlink channels. However, as downlink SSB signals cannot directly reflect the uplink channel conditions, the optimal performance may not be achieved in different channel conditions or grip conditions. To address this issue, closed-loop PUSCH antenna selection is introduced. A closed-loop mechanism enables the base station to instruct UEs to select antennas for uplink transmission based on uplink SRS measurement results, which can directly reflect the uplink channel conditions. In this way, a UE can select the antennas that deliver better performance.

After closed-loop PUSCH antenna selection is enabled, **the base station measures the SRS RSRP values of different antennas of a UE that supports this function. It then instructs the UE to select the antennas with higher SRS RSRP for data transmission over the PUSCH**, increasing the uplink UE throughput. **4.1.10 Closed-Loop PUSCH Antenna Selection** uses a 1T4R UE as an example to describe the principles of closed-loop PUSCH antenna selection. This function is controlled by the **PUSCH_CLOSED_LOOP_ANT_SEL_SW** option of the **NRDUCellChnCovAlgo.UlCoverageAlgoSwitch** parameter.

It is recommended that this function be enabled when a large number of UEs support closed-loop PUSCH antenna selection.

Figure 4-3 Principles of closed-loop PUSCH antenna selection



4.2 Network Analysis

4.2.1 Benefits

To maximize the gains of the Turbo Uplink feature, it is recommended that all the following conditions be met before this feature is enabled:

- There are UEs in weak coverage areas. Specifically, the average number of active UEs at the uplink coverage edges in a cell (**N.User.RRConn.Active.UL.Edge.Avg**) per hour in a day is greater than or equal to 0.5, and the proportion of such UEs (**N.User.RRConn.Active.UL.Edge.Avg/N.User.RRConn.Active.UL.Avg × 100%**) is greater than 5%.
- The average uplink interference (**N.UL.NI.Avg**) is greater than or equal to -110 dBm.
- The uplink PRB usage (**Uplink Resource Block Utilizing Rate (DU)**) is less than 40%.

Turbo Uplink increases the uplink user-perceived rate at the coverage edge and the **User Uplink Average Throughput (DU)**. The benefits of each function are as follows:

- RRC-reconfiguration-based dynamic waveform switching
 - DFT-s-OFDM delivers a higher uplink user-perceived rate at the coverage edge (**(N.ThpVol.UL.Edge - N.ThpVol.UE.UL.SmallPkt.Edge)/**

N.ThpTime.UE.UL.RmvSmallPkt.Edge) than CP-OFDM under the same scheduling bandwidth conditions in low-frequency scenarios. This is because DFT-s-OFDM allows a higher transmit power of UEs than CP-OFDM. The increases in the transmit power of UEs vary with UEs.

- DFT-s-OFDM delivers 0–2 dB higher UE transmit power than CP-OFDM according to the definition of MPR in section 6.2.2 "UE maximum output power reduction" in 3GPP TS 38.101-1 V15.5.0. The increases in UE transmit power on live networks vary with UEs.
- The MPR values of UEs provided by different vendors meet specifications as long as the values are less than or equal to the protocol-defined ones. Therefore, the transmit power of different UEs differs, and the benefits offered by this function also vary.
- This function does not produce any gains in single-UE multi-stream transmission scenarios, where DFT-s-OFDM is not applicable and the UE still works with CP-OFDM.
- Specific absorption rate (SAR) is a measure of the rate at which energy is absorbed per unit mass by a human body when exposed to an electromagnetic field. A larger value indicates a greater impact on the human body. To ensure safety, the transmit power of a UE must be lower than a specified SAR when the UE is close to the human body. Therefore, to achieve the function gain, it is required that the transmit power of the UE not be limited due to the specified SAR.
- Frequency selective scheduling under weak uplink coverage
 - In a lightly loaded cell (where the average number of active UEs in the uplink, indicated by **N.User.RRConn.Active.UL.Avg**, is small and the uplink PRB usage, indicated by the **Uplink Resource Block Utilizing Rate (DU)**, is less than or equal to 40%), this function increases the uplink user-perceived rate at the coverage edge. The more significant the frequency selective characteristics, the higher the benefits.
 - In a heavily loaded cell (where the average number of active UEs in the uplink (**N.User.RRConn.Active.UL.Avg**) is large or the uplink PRB usage (**Uplink Resource Block Utilizing Rate (DU)**) is greater than 40%), this function does not take effect and therefore has no negative impacts.
- Spatial channel estimation under weak uplink coverage
 - Under light loads, this function increases the uplink user-perceived rate at the coverage edge. The higher the interference, the higher the gain.
 - Under heavy loads, the proportion of UEs for which this function takes effect in weak coverage areas decreases, but there is no negative impact.
- SR-based scheduling optimization

Under uplink heavy loads, this function increases the uplink user-perceived rate at the coverage edge and the **Cell Uplink Average Throughput (DU)**. The larger the number of RRC_CONNECTED UEs or the higher the **Uplink Resource Block Utilizing Rate (DU)**, the higher the gains.

Under uplink light loads, there is no negative impact.
- Uplink SINR filtering optimization

This function increases the uplink user-perceived rate of UEs moving at the coverage edge and requiring a long time to complete SRS frequency hopping over the entire bandwidth.

- Full-power dual transmission
 - Full-power dual transmission increases the uplink transmit power and uplink user-perceived rate of cell-edge UEs that support full-power transmission in the uplink and work in rank 1 in SA networking.
 - To ensure safety, the transmit power of a UE must be lower than a specified SAR when the UE is close to the human body. Therefore, to achieve the function gain, it is required that the transmit power of the UE not be limited due to the specified SAR.
- Uplink $\pi/2$ -BPSK modulation/Uplink $\pi/2$ -BPSK modulation power boost/
Uplink low-PAPR DMRS

To achieve the optimal gains, you are advised to enable all these functions. The gains provided by these functions are as follows:

- If the uplink PRB usage (**Uplink Resource Block Utilizing Rate (DU)**) is less than or equal to 30%, the uplink user-perceived rate at the coverage edge and the **Cell Uplink Average Throughput (DU)** increase. If the uplink PRB usage is greater than 30%, the gains decrease as the uplink PRB usage increases. Specifically, the gain in the uplink user-perceived rate at the coverage edge decreases and may drop to zero, while the gain in the **Cell Uplink Average Throughput (DU)** decreases and may even become negative.
- To ensure safety, the transmit power of a UE must be lower than a specified SAR when the UE is close to the human body. To achieve the function gain, it is required that the transmit power of the UE not be limited due to the specified SAR.
- Closed-loop PUSCH antenna selection
 - For UEs that support and perform closed-loop PUSCH antenna selection, more RBs are allocated to cell-edge UEs. This may increase the **Uplink Resource Block Utilizing Rate (DU)** and also increases the average uplink MCS index, uplink user-perceived rate at the coverage edge, and **Cell Uplink Average Throughput (DU)**.
 - For UEs that do not support closed-loop PUSCH antenna selection, there are no negative impacts.

NOTE

Average uplink MCS index = $(0 \times \text{N.ChMeas.PUSCH.MCS.0} + 1 \times \text{N.ChMeas.PUSCH.MCS.1} + 2 \times \text{N.ChMeas.PUSCH.MCS.2} + \dots + 28 \times \text{N.ChMeas.PUSCH.MCS.28}) / (\text{N.ChMeas.PUSCH.MCS.0} + \text{N.ChMeas.PUSCH.MCS.1} + \text{N.ChMeas.PUSCH.MCS.2} + \dots + \text{N.ChMeas.PUSCH.MCS.28})$

Uplink user-perceived rate at the coverage edge = $(\text{N.ThpVol.UL.Edge} - \text{N.ThpVol.UE.UL.SmallPkt.Edge}) / \text{N.ThpTime.UE.UL.RmvSmallPkt.Edge}$

4.2.2 Impacts

The impacts between this function and other functions are described in [4.3.2.3 Function Impacts](#).

- RRC-reconfiguration-based dynamic waveform switching
 - CP-OFDM can use non-contiguous frequency-domain resources while DFT-s-OFDM can use only contiguous frequency-domain resources. If UEs select DFT-s-OFDM rather than CP-OFDM in non-contiguous frequency-domain scenarios, the scheduling bandwidth is smaller, and the single-UE

throughput may not increase or may decrease. The specific negative impacts depend on the changes in scheduling bandwidth and power spectral density as well as the increase in the transmit power of UEs after switching from CP-OFDM to DFT-s-OFDM.

- After this function is enabled, the maximum transmit power of a UE changes abruptly when the UE switches between CP-OFDM and DFT-s-OFDM. The UE throughput increases significantly when the UE switches from CP-OFDM to DFT-s-OFDM, and decreases significantly when the UE switches from DFT-s-OFDM to CP-OFDM.
- After this function is enabled, compatibility issues may occur on UEs not supporting DFT-s-OFDM, causing service drops.
- After this function is enabled, service drops may occur on lost UEs in a cell. In this case, lost-UE protection against service drops can be used to reduce the number of RRC reconfiguration messages for uplink waveform switching, decreasing the service drop rate in the cell.
- After this function is enabled, the value of the following indicator may fluctuate, depending on the distribution of UEs (in the cell center or at the cell edge):
$$\text{SgNB-initiated SgNB modification success rate in NSA DC scenarios} = \frac{\text{N.NsaDc.SgNB.Mod.Required.Succ}}{\text{N.NsaDc.SgNB.Mod.Required.Att}} \times 100\%$$
- Lost-UE protection against service drops
After this function is enabled, the proportion of UEs using DFT-s-OFDM changes because the number of RRC reconfiguration messages for uplink waveform switching is reduced. As a result, the uplink user-perceived rate at the coverage edge may change.
- Frequency selective scheduling under weak uplink coverage
None
- Spatial channel estimation under weak uplink coverage
After this function takes effect, the uplink MCS index increases. The RBLER may increase in positions where the interference is very high and the IBLER is abnormally high, but the RBLER is almost not affected in the entire cell.
- SR-based scheduling optimization
 - The priorities of UEs with non-zero buffer size increase, and the **Uplink Resource Block Utilizing Rate (DU)** may increase.
 - UEs with non-zero buffer size are scheduled as soon as possible, reducing the number of times SR-based scheduling is performed. The average uplink MCS index may increase.
 - User data is scheduled more quickly, reducing the number of BSRs. The average number of active UEs in a cell may decrease under the same RLC data volume.
 - The number of downlink status reports increases. When downlink PDSCH resources are insufficient, the **Cell Downlink Average Throughput (DU)** and **User Downlink Average Throughput (DU)** decrease.
- Uplink SINR filtering optimization
This function improves the uplink MCS accuracy, decreases the uplink BLER, and increases the uplink user-perceived rate for moving UEs that require a long time to complete SRS frequency hopping over the entire bandwidth.

- Full-power dual transmission
After full-power dual transmission takes effect, the transmit power of cell-edge UEs increases, slightly increasing interference to neighboring cells.
- Uplink $\pi/2$ -BPSK modulation
After this function takes effect, the transmit power of cell-edge UEs increases, slightly increasing interference.
- Uplink $\pi/2$ -BPSK modulation power boost
 - If the uplink cell PRB usage (**Uplink Resource Block Utilizing Rate (DU)**) is greater than 30%, the gain in the **Cell Uplink Average Throughput (DU)** decreases or even drops below zero. This is because more RBs are used by cell-edge UEs for uplink transmission after this function takes effect.
 - The average uplink MCS index decreases.
 - The uplink cell PRB usage increases.
- Uplink low-PAPR DMRS
After this function takes effect, the transmit power of cell-edge UEs increases, slightly increasing interference.
- Closed-loop PUSCH antenna selection
For UEs that support but do not perform closed-loop PUSCH antenna selection, a possible increase in the uplink bit errors may decrease the uplink user-perceived rate at the coverage edge.

4.3 Requirements

4.3.1 Licenses

Feature ID	Feature Name	Model	NE	Sales Unit
FOFD-061205	Turbo Uplink	NR0S00TBUL00	gNodeB	Per Cell

NOTE

The insufficiency of certain feature licenses affects cell activation. For details, see the **Is Cell Activation Affected by License Insufficiency** column in *License Control Item Lists*.

4.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

4.3.2.1 Prerequisite Functions

- RRC-reconfiguration-based dynamic waveform switching

Function Name	Function Switch	Reference
Turbo Uplink Switch	TURBO_UPLINK_SW option of the NRDUCellChnCovAlgo.UlCoverageAlgoSwitch parameter	<i>Turbo Uplink (Low-Frequency TDD)</i>

- Frequency selective scheduling under weak uplink coverage

Function Name	Function Switch	Reference
Turbo Uplink Switch	TURBO_UPLINK_SW option of the NRDUCellChnCovAlgo.UlCoverageAlgoSwitch parameter	<i>Turbo Uplink (Low-Frequency TDD)</i>
Uplink scheduling enhancement	UL_SCHEDULE_ENH_SW option of the NRDUCellPusch.UlPuschAlgoSwitch parameter	<i>MIMO (TDD)</i>
UL Interference Random Schedule Enh Sw	UL_INTRF_RANDOMIZED_SCH_ENH_SW option of the NRDUCellPusch.UlPuschAlgoSwitch parameter	None
Optimal Paired RB Selection Switch	OPT_PAIR_RB_SEL_SW option of the NRDUCellUIMimo.UlMimoAlgoSwitch parameter	None

- Spatial channel estimation under weak uplink coverage

Function Name	Function Switch	Reference	Description
Turbo Uplink Switch	TURBO_UPLINK_SW option of the NRDUCellChnCovAlgo.UlCoverageAlgoSwitch parameter	<i>Turbo Uplink (Low-Frequency TDD)</i>	Spatial channel estimation under weak uplink coverage requires that the Turbo Uplink Switch be turned on.

- SR-based scheduling optimization

Function Name	Function Switch	Reference
SR-based smart scheduling	SR_SMART_SCH_SW option of the NRDUCellPusch.UlPuschAlgoSwitch parameter	<i>Uplink Scheduling</i>
Turbo Uplink Switch	TURBO_UPLINK_SW option of the NRDUCellChnCovAlgo.UlCoverageAlgoSwitch parameter	<i>Turbo Uplink (Low-Frequency TDD)</i>

- Uplink SINR filtering optimization

Function Name	Function Switch	Reference
Turbo Uplink Switch	TURBO_UPLINK_SW option of the NRDUCellChnCovAlgo.UlCoverageAlgoSwitch parameter	<i>Turbo Uplink (Low-Frequency TDD)</i>

- Full-power dual transmission

Function Name	Function Switch	Reference	Description
Turbo Uplink Switch	TURBO_UPLINK_SW option of the NRDUCellChnCovAlgo.UlCoverageAlgoSwitch parameter	<i>Turbo Uplink (Low-Frequency TDD)</i>	Full-power dual transmission requires that the Turbo Uplink Switch be turned on.

- Uplink $\pi/2$ -BPSK modulation

Function Name	Function Switch	Reference
RRC-reconfiguration-based dynamic waveform switching	UL_WAVEFORM_ADAPT_SW option of the NRDUCellAlgoSwitch.AdaptiveEdgeExpEnhSwitch parameter	<i>Turbo Uplink (Low-Frequency TDD)</i>

Function Name	Function Switch	Reference
Turbo Uplink Switch	TURBO_UPLINK_SW option of the NRDUCellChnCovAlgo.UlCoverageAlgoSwitch parameter	<i>Turbo Uplink (Low-Frequency TDD)</i>

- Uplink $\pi/2$ -BPSK modulation power boost

Function Name	Function Switch	Reference
Uplink $\pi/2$ -BPSK modulation	UL_HALF_PI_BPSK_SW option of the NRDUCellAlgoSwitch.AdaptiveEdgeExpEnhSwitch parameter	<i>Turbo Uplink (Low-Frequency TDD)</i>
Cell frequency band of n40/n41/n77/n78/n79	NRDUCell.FrequencyBand and set to N40, N41, N77, N78, or N79	<i>Cell Management</i>
Turbo Uplink Switch	TURBO_UPLINK_SW option of the NRDUCellChnCovAlgo.UlCoverageAlgoSwitch parameter	<i>Turbo Uplink (Low-Frequency TDD)</i>

- Uplink low-PAPR DMRS

Function Name	Function Switch	Reference
Uplink $\pi/2$ -BPSK modulation power boost	UL_HALF_PI_BPSK_POWER_BOOST_SW option of the NRDUCellAlgoSwitch.AdaptiveEdgeExpEnhSwitch parameter	<i>Turbo Uplink (Low-Frequency TDD)</i>
Turbo Uplink Switch	TURBO_UPLINK_SW option of the NRDUCellChnCovAlgo.UlCoverageAlgoSwitch parameter	<i>Turbo Uplink (Low-Frequency TDD)</i>

- Closed-loop PUSCH antenna selection

Function Name	Function Switch	Reference	Description
Turbo Uplink Switch	TURBO_UPLINK_SW option of the NRDUCellChnCo vAlgo.UlCoverageAlgoSwitch parameter	<i>Turbo Uplink (Low-Frequency TDD)</i>	Closed-loop PUSCH antenna selection requires that the Turbo Uplink Switch be turned on.
Frequency Band	NRDUCell.FrequencyBand	None	Closed-loop PUSCH antenna selection requires that NRDUCell.FrequencyBand be set to N38, N40, N41, N77, N78, or N79 .

4.3.2.2 Mutually Exclusive Functions

- RRC-reconfiguration-based dynamic waveform switching
None
- Frequency selective scheduling under weak uplink coverage

Function Name	Function Switch	Reference	Description
Frequency Band	NRDUCell.FrequencyBand	None	Frequency selective scheduling under weak uplink coverage cannot be enabled if NRDUCell.FrequencyBand is set to N257, N258, N260, or N261 .

- Spatial channel estimation under weak uplink coverage

Function Name	Function Switch	Reference	Description
High-speed Railway Superior Experience	NRDUCell.HighSpeedFlag set to HIGH_SPEED	<i>High Speed Mobility</i>	Spatial channel estimation under weak uplink coverage cannot be enabled for high-speed cells.

- SR-based scheduling optimization
None
- Uplink SINR filtering optimization
None
- Full-power dual transmission
None
- Uplink $\pi/2$ -BPSK modulation

Function Name	Function Switch	Reference	Description
UL 256QAM	NRDUCellAlgoSwitch.Ul256QamSwitch set to UL_256QAM_FIXED	<i>Modulation Schemes</i>	Uplink $\pi/2$ -BPSK modulation cannot be enabled if NRDUCellAlgoSwitch.Ul256QamSwitch is set to UL_256QAM_FIXED .
Frequency Band	NRDUCell.FrequencyBand	None	Uplink $\pi/2$ -BPSK modulation cannot be enabled if NRDUCell.FrequencyBand is set to N257 , N258 , N260 , or N261 .

- Uplink $\pi/2$ -BPSK modulation power boost
None
- Uplink low-PAPR DMRS

Function Name	Function Switch	Reference	Description
Distributed massive MIMO	DM_MIMO_SERVICE_SWITCH option of the NRDUCellAlgoSwitch.DmMimoSwitch parameter	<i>Distributed Antenna Solutions (Low-Frequency TDD)</i>	If uplink low-PAPR DMRS is enabled, distributed massive MIMO cannot be enabled.

- Closed-loop PUSCH antenna selection

Function Name	Function Switch	Reference	Description
High-speed Railway Superior Experience	NRDUCell.HighSpeedFlag set to HIGH_SPEED	<i>High Speed Mobility</i>	Closed-loop PUSCH antenna selection and High-speed Railway Superior Experience cannot be both enabled.
Hyper Cell	NRDUCell.NrDuCellNetworkingMode set to HYPER_CELL	<i>Hyper Cell</i>	Closed-loop PUSCH antenna selection cannot be enabled for hyper cells.
Cell Combination	NRDUCell.NrDuCellNetworkingMode set to HYPER_CELL_COMBINE_MODE	<i>Cell Combination</i>	Closed-loop PUSCH antenna selection cannot be enabled for combined cells.
Fusion Cell	NRDUCell.NrDuCellNetworkingMode set to FUSION_CELL	<i>Fusion Cell (Low-Frequency TDD)</i>	Closed-loop PUSCH antenna selection and Fusion Cell cannot be both enabled.

4.3.2.3 Function Impacts

- RRC-reconfiguration-based dynamic waveform switching

Function Name	Function Switch	Reference	Description
Cell Combination	NRDUCell.NrDuCellNetworkingMode set to HYPER_CELL_COMBINE_MODE	<i>Cell Combination</i>	RRC-reconfiguration-based dynamic waveform switching does not take effect for a combined cell in low-speed scenarios.
Joint Transmission Cell Combination	• NRDUCell.NrDuCellNetworkingMode set to HYPER_CELL_COMBINE_MODE • NRDUCellMultiTrp.DataTransMode set to JOINT_MODE	<i>Cell Combination</i>	RRC-reconfiguration-based dynamic waveform switching does not take effect for a combined cell in joint transmission mode in low-speed scenarios.

Function Name	Function Switch	Reference	Description
Hyper Cell	NRDUCell.NrDuCellNe tworkingMode set to HYPER_CELL	<i>Hyper Cell</i>	RRC-reconfiguration- based dynamic waveform switching does not take effect for a hyper cell in low- speed scenarios.
High-speed cell	NRDUCell.HighSpeedF lag set to HIGH_SPEED	<i>High Speed Mobility</i>	After RRC- reconfiguration-based dynamic waveform switching is enabled in a high-speed cell, UEs whose uplink capability is rank 1 always use the DFT waveform when they perform full buffer services. You are advised to enable this function when the overall coverage along the high-speed railway is poor (for example, the average SSB RSRP is less than -90 dBm). Enabling this function can increase the uplink throughput of UEs performing full buffer services when the overall coverage along the high-speed railway is poor, but may decrease the uplink throughput of these UEs when the overall coverage along the high-speed railway is good (for example, the average SSB RSRP is greater than -80 dBm).

Function Name	Function Switch	Reference	Description
PUSCH MU-MIMO	UL_MU_MIMO_SW option of the NRDUCellAlgoSwitch . <i>MuMimoSwitch</i> parameter	<i>MIMO (TDD)</i>	If both PUSCH MU-MIMO and RRC-reconfiguration-based dynamic waveform switching are enabled, UEs using the DFT-s-OFDM waveform and those using the CP-OFDM waveform cannot be paired for MU-MIMO. As a result, the MU-MIMO pairing ratio decreases and the gains provided by PUSCH MU-MIMO may decrease.
Uplink waveform control	UL_WAVEFORM_CTRL_SW option of the NRDUCellVoiceAlgo . <i>VoiceUserAlgoSw</i> parameter	<i>VoNR</i>	If both uplink waveform control and RRC-reconfiguration-based dynamic waveform switching are enabled, voice service UEs always use the DFT-s-OFDM waveform for the PUSCH.
DCI-based dynamic waveform switching	DYNAMIC_WAVEFORM_SWITCHING_SW option of the NRDUCellChnCovAlgo . <i>UlCoverageAlgoSwitch</i> parameter	<i>Uplink Scheduling</i>	In the case that both RRC-reconfiguration-based dynamic waveform switching and DCI-based dynamic waveform switching are enabled, if the UE capability meets the requirements of DCI-based dynamic waveform switching (for details, see <i>Uplink Scheduling</i>), DCI-based dynamic waveform switching takes effect; otherwise, RRC-reconfiguration-based dynamic waveform switching takes effect.

- Frequency selective scheduling under weak uplink coverage

None

- Spatial channel estimation under weak uplink coverage

Function Name	Function Switch	Reference	Description
Uplink dual filtering	UL_SRS_DMRS_DUAL_FILTER_SW option of the NRDUCellPusch.UlPuschAlgoSwitch parameter	<i>Uplink Scheduling</i>	When spatial channel estimation under weak uplink coverage is enabled, uplink dual filtering is not recommended. Instead, it is recommended that high-reliability uplink MCS selection be enabled in this case to allow for more accurate MCS selection and higher average uplink throughput of edge UEs. (This function is controlled by the UL_HIGH_RELIABILITY_MCS_SEL_SW option of the NRDUCellUplink.UlAmcPerformanceSw parameter.)

Function Name	Function Switch	Reference	Description
Uplink PAMC	UL_PAMC_SWITCH option of the NRDUCellPusch.UlPuschAlgoSwitch parameter	<i>Uplink Scheduling</i>	When spatial channel estimation under weak uplink coverage is enabled, uplink PAMC is not recommended. Instead, it is recommended that probability-based uplink MCS selection (controlled by the UL_PBMCSSEL_SW option of the NRDUCellUplink.UlAmcPerformanceSw parameter) be enabled in this case to allow for more accurate MCS selection and higher average uplink throughput of cell-edge UEs.
Uplink 4-layer transmission for a single FWA user	UL_SU_FOUR_LAYER_SW option of the NRDUCellUplink.UlRankAlgorithm parameter	<i>FWA</i>	If uplink 4-layer transmission for a single FWA user has taken effect, spatial channel estimation under weak uplink coverage does not take effect for the user.

- SR-based scheduling optimization
None
- Uplink SINR filtering optimization
None

- Full-power dual transmission

Function Name	Function Switch	Reference	Description
Coherent-codebook-based scheduling	COHERENT_CODEBOOK_SCH_SW option of the NRDUCellChnConfigAlgo.UlCoverageAlgoSwitch parameter	<i>Channel Management</i>	If both full-power dual transmission and coherent-codebook-based scheduling are enabled, for UEs that support coherent-codebook-based scheduling and only the fullpowerMode1 transmission mode of full-power dual transmission, coherent-codebook-based scheduling preferentially takes effect.
Uplink 4-layer transmission for a single FWA user	UL_SU_FOUR_LAYER_SW option of the NRDUCellUIRankAlgoSw parameter	<i>FWA</i>	If uplink 4-layer transmission for a single FWA user has taken effect, full-power dual transmission does not take effect for the user.

- Uplink $\pi/2$ -BPSK modulation

Function Name	Function Switch	Reference	Description
Msg3 transform precoding configuration	MSG3_TRANSFORMPRECODING_SWITCH option of the NRDUCellPusch.UlPuschAlgoSwitch parameter	None	Uplink $\pi/2$ -BPSK modulation is not recommended when Msg3 transform precoding configuration is enabled. In the case that both functions are enabled, if a UE performs contention-based access (in scenarios such as handover to the LTE side in NSA networking and resynchronization) after it initially accesses the network, indicators related to contention-based access and service drops deteriorate because the UE and the base station use different modulation schemes.

- Uplink $\pi/2$ -BPSK modulation power boost
None
- Uplink low-PAPR DMRS
None
- Closed-loop PUSCH antenna selection

Function Name	Function Switch	Reference	Description
Downlink 8-layer transmission for a single FWA user	DL_SU_EIGHT_LAYER_SW option of the NRDUCellDLRank.DlRankSelAlgorithm parameter	<i>FWA</i>	If downlink 8-layer transmission for a single FWA user has taken effect, closed-loop PUSCH antenna selection does not take effect for the user.

Function Name	Function Switch	Reference	Description
Intelligent uplink multi-layer SINR prediction	UL_SU_SINR_INTEL_PREDICT_SW option of the NRDUCellAiAlgo.AiAmcAlgoSwitch parameter	<i>MIMO (TDD)</i>	Closed-loop PUSCH antenna selection and intelligent uplink multi-layer SINR prediction cannot take effect at the same time, and the former preferentially takes effect.
Uplink precise MCS optimization	UL_PRECISE_MCS_OPT_SW option of the NRDUCellAiAlgo.AiAmcAlgoSwitch parameter	<i>Massive MIMO Uplink Boosting (Low-Frequency TDD)</i>	Closed-loop PUSCH antenna selection preferentially takes effect when it is enabled together with uplink precise MCS optimization.
Uplink 4-layer transmission for a single FWA user	UL_SU_FOUR_LAYER_SW option of the NRDUCellUlRank.UlRankAlgoSw parameter	<i>FWA</i>	If uplink 4-layer transmission for a single FWA user has taken effect, closed-loop PUSCH antenna selection does not take effect for the user.

4.3.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

Base Station Models

3900 and 5900 series base stations (macro base stations). 3900 series base stations must be configured with the BBU3910.

Boards

- Uplink $\pi/2$ -BPSK modulation/Uplink $\pi/2$ -BPSK modulation power boost/Uplink low-PAPR DMRS

All NR-capable main control boards as well as NR TDD-capable UBBPg and later baseband processing units support this function. For details, see the BBU technical specifications in *3900 & 5900 Series Base Station Product Documentation*.

- Other functions

All NR-capable main control boards and NR TDD-capable baseband processing units support this function. For details, see the BBU technical specifications in *3900 & 5900 Series Base Station Product Documentation*.

RF Modules

- Uplink low-PAPR DMRS/Spatial channel estimation under weak uplink coverage

All NR-capable RF modules with at least 32T32R support this function. For details, see the technical specifications of RF modules in *3900 & 5900 Series Base Station Product Documentation*.

- Other functions

All NR TDD-capable RF modules that work in low frequency bands support this function. For details, see the technical specifications of RF modules in *3900 & 5900 Series Base Station Product Documentation*.

4.3.4 Cells

- Spatial channel estimation under weak uplink coverage
 - The **NRDUCell.FrequencyBand** parameter must be set to **N38, N40, N41, N77, N78, or N79**.
 - The **NRDUCell.SlotAssignment** parameter must be set to **4_1_DDDSU, 8_2_DDDDDDDSUU, or 8_2_DDDSUUDDDD**.
- Uplink low-PAPR DMRS
 - The **NRDUCellTrp.TxRxMode** parameter must be set to **32T32R or 64T64R**.
 - The **NRDUCell.FrequencyBand** parameter must be set to **N40, N41, N77, N78, or N79**.
 - Uplink low-PAPR DMRS cannot be enabled if the **NRDUCell.SlotAssignment** parameter is set to **2_3_DSUUU**.
- Other functions
None

4.3.5 Others

- Full-power dual transmission
 - Full-power dual transmission requires that UEs support at least one of the following transmission modes: fullpower, fullpowerMode2, and fullpowerMode1.
 - Whether a UE supports these transmission modes is determined by the following capability fields in the *FeatureSetUplink* IE: *ul-FullPwrMode-r16*, *ul-FullPwrMode2-TPMIGroup-r16*, and *ul-FullPwrMode1-r16*.
- Uplink $\pi/2$ -BPSK modulation

This function requires that UEs support uplink $\pi/2$ -BPSK modulation. If the *Phy-Parameters* IE reported by a UE contains the *pusch-HalfPi-BPSK* field, the UE supports this function.

- Closed-loop PUSCH antenna selection
This function requires that UEs support closed-loop PUSCH antenna selection. If the value of the *maxNumberSRS-ResourcePerSet* field in the *FeatureSetUplinkPerCC* IE is 2, UEs support this function.
- Other functions
None

4.4 Operation and Maintenance

4.4.1 Data Configuration

4.4.1.1 Data Preparation

[Table 4-1](#), [Table 4-2](#), [Table 4-3](#), [Table 4-4](#), [Table 4-5](#), [Table 4-6](#), [Table 4-7](#), [Table 4-8](#), [Table 4-9](#), [Table 4-10](#), [Table 4-11](#), and [Table 4-12](#) describe the parameters used for function activation. This section does not describe parameters related to cell establishment.

Table 4-1 Parameters that are recommended to be configured before activating Turbo Uplink

Parameter Name	Parameter ID	Option	Setting Notes
Uplink PUSCH Algorithm Switch	NRDUCellPusch. <i>UlPuschAlgoSwitch</i>	UL_SCHEDULE_ENH_SW	To maximize the gains of Turbo Uplink, it is recommended that this option be selected before this feature is enabled.
Uplink PUSCH Algorithm Switch	NRDUCellPusch. <i>UlPuschAlgoSwitch</i>	UL_INTRF_RANDOMIZED_SCH_ENH_SW	To maximize the gains of Turbo Uplink, it is recommended that this option be selected before this feature is enabled.

Parameter Name	Parameter ID	Option	Setting Notes
UL MU-MIMO Algorithm Switch	NRDUCellULMimo.<i>o.ULMuMimoAlgoSwitch</i>	OPT_PAIR_RB_SEL_SW	To maximize the gains of Turbo Uplink, it is recommended that this option be selected before this feature is enabled.
PUSCH Performance Switch	NRDUCellPusch.<i>PuschPerformanceSwitch</i>	UL_FREQ_SEL_SCH_OPT_SW	To maximize the gains of Turbo Uplink, it is recommended that this option be selected before this feature is enabled.
UL MU-MIMO Algorithm Switch	NRDUCellULMimo.<i>o.ULMuMimoAlgoSwitch</i>	BIG_PKT_ALIGN_PAIR_OPT_SW	To maximize the gains of Turbo Uplink, it is recommended that this option be selected before this feature is enabled.
UL MU-MIMO Algorithm Switch	NRDUCellULMimo.<i>o.ULMuMimoAlgoSwitch</i>	SML_PKT_ALIGN_PAIR_OPT_SW	To maximize the gains of Turbo Uplink, it is recommended that this option be selected before this feature is enabled.
Uplink AMC Algorithm Switch	NRDUCellULAmc.<i>ULAmcAlgoSw</i>	MU_AMC_OPT_SW	To maximize the gains of Turbo Uplink, it is recommended that this option be selected before this feature is enabled.

Parameter Name	Parameter ID	Option	Setting Notes
Uplink AMC Algorithm Switch	NRDUCellULAmc. <i>ULAmcAlgoSw</i>	UL_SCH_AVG_SINR_OPT_SW	To maximize the gains of Turbo Uplink, it is recommended that this option be selected before this feature is enabled.
Uplink PUSCH Algorithm Switch	NRDUCellPusch. <i>ULPuschAlgoSwitch</i>	SR_SMART_SCH_SW	To maximize the gains of Turbo Uplink, it is recommended that this option be selected before this feature is enabled.
PUSCH Performance Switch	NRDUCellPusch. <i>PuschPerformanceSwitch</i>	UL_NONCONSEC_SCH_OPT_SW	To maximize the gains of Turbo Uplink, it is recommended that this option be selected before this feature is enabled.
MU RB Ratio Threshold	NRDUCellULMimo. <i>MuRbRatioThld</i>	None	Set this parameter based on the network plan. To maximize the gains of Turbo Uplink, it is recommended that this parameter be set to 40 before this feature is enabled.
UL MU-MIMO SINR Threshold	NRDUCellULMimo. <i>ULMuMimoSinrThld</i>	None	To maximize the gains of Turbo Uplink, it is recommended that this parameter be set to -20 before this feature is enabled.

Parameter Name	Parameter ID	Option	Setting Notes
Uplink 256QAM Switch	NRDUCellAlgoSwitch.Ul256QamSwitch	None	To maximize the gains of Turbo Uplink, it is recommended that this parameter be set to UL_256QAM_ADAPT before this feature is enabled.
Uplink AMC Algorithm Switch	NRDUCellUlAmc.UlAmcAlgoSwitch	UL_256QAM_ADAPT_POWER_OPT_SW	To maximize the gains of Turbo Uplink, it is recommended that this option be selected before this feature is enabled.

Table 4-2 Parameters used for activating Turbo Uplink

Parameter Name	Parameter ID	Option	Setting Notes
UL Coverage Algorithm Switch	NRDUCellChnCoverageAlgo.UlCoverageAlgoSwitch	TURBO_UPLINK_SW	Select this option if the Turbo Uplink feature is required.

Table 4-3 Parameters used for activating RRC-reconfiguration-based dynamic waveform switching

Parameter Name	Parameter ID	Option	Setting Notes
Adaptive Edge Experience Enhance Switch	NRDUCellAlgoSwitch.AdaptiveEdgeExpEnhSwitch	UL_WAVEFORM_ADAPT_SW	Select this option. It is recommended that the UL_WAVEFORM_ADAPT_SW_OFF option of the gNBUECompatibility.BlacklistCtrlSwitch parameter also be selected to resolve compatibility issues of some UEs.
SINR Threshold for Waveform Selection	NRDUCellPusch.SinrThldforWaveformSel	None	Retain the default value.
Uplink AMC Performance Switch	NRDUCellULAmc.UlAmcPerformanceSw	LOST_UE_DROP_PROTECT_SW	Select this option.

Table 4-4 Parameters used for activating frequency selective scheduling under weak uplink coverage

Parameter Name	Parameter ID	Option	Setting Notes
UL Coverage Algorithm Switch	NRDUCellChnCovAlgo.UlCoverageAlgoSwitch	UL_WEAK_COV_FREQ_SEL_SCH_SW	Select this option.

Table 4-5 Parameters used for activating spatial channel estimation under weak uplink coverage

Parameter Name	Parameter ID	Option	Setting Notes
UL Coverage Algorithm Switch	NRDUCellChnCovAlgo.UlCoverageAlgoSwitch	UL_WEAK_COV_SPACE_CHN_ESTI_SW	Select this option.

Table 4-6 Parameters used for activating SR-based scheduling optimization

Parameter Name	Parameter ID	Option	Setting Notes
PUSCH Performance Switch	NRDUCellPusch.<i>PuschPerformanceSwitch</i>	SR_SCH_OPT_SW	Select this option.

Table 4-7 Parameters used for activating uplink SINR filtering optimization

Parameter Name	Parameter ID	Option	Setting Notes
Uplink AMC Algorithm Switch	NRDUCellULAmc.<i>ULAmcAlgoSw</i>	UL_SINR_FILTER_OPT_SW	Select this option.

Table 4-8 Parameters used for activating full-power dual transmission

Parameter Name	Parameter ID	Option	Setting Notes
Adaptive Edge Experience Enhance Switch	NRDUCellAlgoSwitch.<i>AdaptiveEdgeExpEnhSwitch</i>	UL_RANK1_FULL_POWER_SW	It is recommended that this option be selected in scenarios with limited coverage and be deselected in other scenarios.

Table 4-9 Parameters used for activating uplink $\pi/2$ -BPSK modulation

Parameter Name	Parameter ID	Option	Setting Notes
Adaptive Edge Experience Enhance Switch	NRDUCellAlgoSwitch.AdaptiveEdgeExpEnhSwitch	UL_HALF_PI_BPSK_SW	It is recommended that this option be selected in scenarios with limited coverage and be deselected in other scenarios. If the MSG3_TRANSFORMPRECODE_SWITCH option of the NRDUCellPusch.UlPuschAlgoSwitch parameter is selected, uplink $\pi/2$ -BPSK modulation is not recommended.

Table 4-10 Parameters used for activating uplink $\pi/2$ -BPSK modulation power boost

Parameter Name	Parameter ID	Option	Setting Notes
Adaptive Edge Experience Enhance Switch	NRDUCellAlgoSwitch.AdaptiveEdgeExpEnhSwitch	UL_HALF_PI_BPSK_POWER_BOOST_SW	If the uplink PRB usage is less than 30%, selecting this option is recommended. Otherwise, deselecting this option is recommended.
Half-Pi BPSK Power Boost Value	NRDUCellUlamc.HalfPiBpskPwrBoostValue	None	Retain the default value.
RB Limit SINR Thld in Power Boost	NRDUCellUlamc.PwrBoostRbLimitSinrThld	None	Retain the default value.
RB Limit SINR Thld Offset in Power Boost	NRDUCellUlamc.PwrBoostRbLimitSinrThldOfs	None	Retain the default value.

Table 4-11 Parameters used for activating uplink low-PAPR DMRS

Parameter Name	Parameter ID	Option	Setting Notes
Adaptive Edge Experience Enhance Switch	NRDUCellAlgoSwitch.AdaptiveEdgeExpEnhSwitch	UL_LOW_PAPR_DMRS_SW	Select this option.

Table 4-12 Parameters used for activating closed-loop PUSCH antenna selection

Parameter Name	Parameter ID	Option	Setting Notes
UL Coverage Algorithm Switch	NRDUCellChnCoverageAlgoSwitch	PUSCH_CLOSED_LOOP_ANT_SEL_SW	Select this option if closed-loop PUSCH antenna selection is required.

4.4.1.2 Using MML Commands

Before using MML commands, refer to [4.3 Requirements](#) and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Activation Command Examples

- To maximize the gains provided by this feature, it is recommended that the following operations be performed before this feature is enabled:

```
//Enabling uplink scheduling enhancement
MOD NRDUCCELLPUSCH: NrDuCellId=0, UlPuschAlgoSwitch=UL_SCHEDULE_ENH_SW-1;
//Enabling uplink interference randomization-based scheduling enhancement
MOD NRDUCCELLPUSCH: NrDuCellId=0,
UlPuschAlgoSwitch=UL_INTRF_RANDOMIZED_SCH_ENH_SW-1;
//Enabling optimal paired RB selection
MOD NRDUCCELLULMIMO: NrDuCellId=0, UlMuMimoAlgoSwitch=OPT_PAIR_RB_SEL_SW-1;
//Enabling uplink frequency selective scheduling optimization
MOD NRDUCCELLPUSCH: NrDuCellId=0, PuschPerformanceSwitch=UL_FREQ_SEL_SCH_OPT_SW-1;
//Enabling alignment pairing optimization for large-packet UEs
MOD NRDUCCELLULMIMO: NrDuCellId=0, UlMuMimoAlgoSwitch=BIG_PKT_ALIGN_PAIR_OPT_SW-1;
//Enabling alignment pairing optimization for small-packet UEs
MOD NRDUCCELLULMIMO: NrDuCellId=0, UlMuMimoAlgoSwitch=SML_PKT_ALIGN_PAIR_OPT_SW-1;
//Selecting the MU_AMC_OPT_SW option
MOD NRDUCCELLULAMC: NrDuCellId=0, UlAmcAlgoSw=MU_AMC_OPT_SW-1;
//Enabling average SINR optimization in uplink scheduling
MOD NRDUCCELLULAMC: NrDuCellId=0, UlAmcAlgoSw=UL_SCH_AVG_SINR_OPT_SW-1;
//Enabling SR-based smart scheduling
MOD NRDUCCELLPUSCH: NrDuCellId=0, UlPuschAlgoSwitch=SR_SMART_SCH_SW-1;
//Enabling uplink non-consecutive scheduling optimization
MOD NRDUCCELLPUSCH: NrDuCellId=0, PuschPerformanceSwitch=UL_NONCONSEC_SCH_OPT_SW-1;
```

- ```
//Setting the MuRbRatioThld parameter (not required if this parameter has been set to 60 on the live network)
MOD NRDUCELLULMIMO: NrDuCellId=0, MuRbRatioThld=40;
//Setting the ULMuMimoSinrThld parameter
MOD NRDUCELLULMIMO: NrDuCellId=0, ULMuMimoSinrThld=-20;
//Setting the UL256QamSwitch parameter
MOD NRDUCELLALGOSWITCH: NrDuCellId=0, UL256QamSwitch=UL_256QAM_ADAPT;
//Selecting the UL_256QAM_ADAPT_POWER_OPT_SW option
MOD NRDUCELLULAMC: NrDuCellId=0, UIAmcAlgoSw=UL_256QAM_ADAPT_POWER_OPT_SW-1;
```
- **Turbo Uplink**  
//Enabling the Turbo Uplink feature  
MOD NRDUCELLCHNCOVALGO: NrDuCellId=0, UICoverageAlgoSwitch=TURBO\_UPLINK\_SW-1;
  - **RRC-reconfiguration-based dynamic waveform switching**  
//Enabling RRC-reconfiguration-based dynamic waveform switching  
MOD NRDUCELLALGOSWITCH: NrDuCellId=0, AdaptiveEdgeExpEnhSwitch=UL\_WAVEFORM\_ADAPT\_SW-1;  
//Setting the SINR threshold for waveform selection  
MOD NRDUCELLPUSCH: NrDuCellId=0, SinrThldforWaveformSel=32;  
//Enabling lost-UE protection against service drops  
MOD NRDUCELLULAMC: NrDuCellId=0, UIAmcPerformanceSw=LOST\_UE\_DROP\_PROTECT\_SW-1;
  - **Frequency selective scheduling under weak uplink coverage**  
//Enabling frequency selective scheduling under weak uplink coverage  
MOD NRDUCELLCHNCOVALGO: NrDuCellId=0, UICoverageAlgoSwitch=UL\_WEAK\_COV\_FREQ\_SEL\_SCH\_SW-1;
  - **Spatial channel estimation under weak uplink coverage**  
//Enabling spatial channel estimation under weak uplink coverage  
MOD NRDUCELLCHNCOVALGO: NrDuCellId=0, UICoverageAlgoSwitch=UL\_WEAK\_COV\_SPACE\_CHN\_ESTI\_SW-1;
  - **SR-based scheduling optimization**  
//Enabling SR-based scheduling optimization  
MOD NRDUCELLPUSCH: NrDuCellId=0, PuschPerformanceSwitch=SR\_SCH\_OPT\_SW-1;
  - **Uplink SINR filtering optimization**  
//Enabling uplink SINR filtering optimization  
MOD NRDUCELLULAMC: NrDuCellId=0, UIAmcAlgoSw=UL\_SINR\_FILTER\_OPT\_SW-1;
  - **Full-power dual transmission**  
//Enabling full-power dual transmission  
MOD NRDUCELLALGOSWITCH: NrDuCellId=0, AdaptiveEdgeExpEnhSwitch=UL\_RANK1\_FULL\_POWER\_SW-1;
  - **Uplink  $\pi/2$ -BPSK modulation**  
//Enabling uplink  $\pi/2$ -BPSK modulation (not recommended if the MSG3\_TRANSFORMPRECODE\_SWITCH option of the NRDUCellPusch.UIPuschAlgoSwitch parameter is selected)  
MOD NRDUCELLALGOSWITCH: NrDuCellId=0, AdaptiveEdgeExpEnhSwitch=UL\_HALF\_PI\_BPSK\_SW-1;
  - **Uplink  $\pi/2$ -BPSK modulation power boost**  
//Selecting the UL\_HALF\_PI\_BPSK\_POWER\_BOOST\_SW option  
MOD NRDUCELLALGOSWITCH: NrDuCellId=0, AdaptiveEdgeExpEnhSwitch=UL\_HALF\_PI\_BPSK\_POWER\_BOOST\_SW-1;  
//Setting the HalfPiBpskPwrBoostValue parameter  
MOD NRDUCELLULAMC: NrDuCellId=0, HalfPiBpskPwrBoostValue=30;  
//Setting the PwrBoostRbLimitSinrThld parameter  
MOD NRDUCELLULAMC: NrDuCellId=0, PwrBoostRbLimitSinrThld=180;  
//Setting the PwrBoostRbLimitSinrThldOfs parameter  
MOD NRDUCELLULAMC: NrDuCellId=0, PwrBoostRbLimitSinrThldOfs=10;
  - **Uplink low-PAPR DMRS**  
//Enabling uplink low-PAPR DMRS  
MOD NRDUCELLALGOSWITCH: NrDuCellId=0, AdaptiveEdgeExpEnhSwitch=UL\_LOW\_PAPR\_DMRS\_SW-1;
  - **Closed-loop PUSCH antenna selection**  
//Enabling closed-loop PUSCH antenna selection  
MOD NRDUCELLCHNCOVALGO: NrDuCellId=0, UICoverageAlgoSwitch=PUSCH\_CLOSED\_LOOP\_ANT\_SEL\_SW-1;

## Deactivation Command Examples

The following provides only deactivation command examples. You can determine whether to restore the settings of other parameters based on actual network conditions.

- **Closed-loop PUSCH antenna selection**  
//Disabling closed-loop PUSCH antenna selection  
MOD NRDUCELLCHNCOVALGO: NrDuCellId=0,  
UICoverageAlgoSwitch=PUSCH\_CLOSED\_LOOP\_ANT\_SEL\_SW-0;
- **Uplink low-PAPR DMRS**  
//Disabling uplink low-PAPR DMRS  
MOD NRDUCELLALGOSWITCH: NrDuCellId=0,  
AdaptiveEdgeExpEnhSwitch=UL\_LOW\_PAPR\_DMRS\_SW-0;
- **Uplink  $\pi/2$ -BPSK modulation power boost**  
//Deselecting the UL\_HALF\_PI\_BPSK\_POWER\_BOOST\_SW option  
MOD NRDUCELLALGOSWITCH: NrDuCellId=0,  
AdaptiveEdgeExpEnhSwitch=UL\_HALF\_PI\_BPSK\_POWER\_BOOST\_SW-0;
- **Uplink  $\pi/2$ -BPSK modulation**  
//Disabling uplink  $\pi/2$ -BPSK modulation  
MOD NRDUCELLALGOSWITCH: NrDuCellId=0, AdaptiveEdgeExpEnhSwitch=UL\_HALF\_PI\_BPSK\_SW-0;
- **Full-power dual transmission**  
//Disabling full-power dual transmission  
MOD NRDUCELLALGOSWITCH: NrDuCellId=0,  
AdaptiveEdgeExpEnhSwitch=UL\_RANK1\_FULL\_POWER\_SW-0;
- **Uplink SINR filtering optimization**  
//Disabling uplink SINR filtering optimization  
MOD NRDUCELLULAMC: NrDuCellId=0, UIAmcAlgoSw=UL\_SINR\_FILTER\_OPT\_SW-0;
- **SR-based scheduling optimization**  
//Disabling SR-based scheduling optimization  
MOD NRDUCELLPUSCH: NrDuCellId=0, PuschPerformanceSwitch=SR\_SCH\_OPT\_SW-0;
- **Spatial channel estimation under weak uplink coverage**  
//Disabling spatial channel estimation under weak uplink coverage  
MOD NRDUCELLCHNCOVALGO: NrDuCellId=0,  
UICoverageAlgoSwitch=UL\_WEAK\_COV\_SPACE\_CHN\_ESTI\_SW-0;
- **Frequency selective scheduling under weak uplink coverage**  
//Disabling frequency selective scheduling under weak uplink coverage  
MOD NRDUCELLCHNCOVALGO: NrDuCellId=0,  
UICoverageAlgoSwitch=UL\_WEAK\_COV\_FREQ\_SEL\_SCH\_SW-0;
- **RRC-reconfiguration-based dynamic waveform switching**  
//Disabling RRC-reconfiguration-based dynamic waveform switching  
MOD NRDUCELLALGOSWITCH: NrDuCellId=0,  
AdaptiveEdgeExpEnhSwitch=UL\_WAVEFORM\_ADAPT\_SW-0;  
//Disabling lost-UE protection against service drops  
MOD NRDUCELLULAMC: NrDuCellId=0, UIAmcPerformanceSw=LOST\_UE\_DROP\_PROTECT\_SW-0;
- **Turbo Uplink**  
//Disabling the Turbo Uplink feature  
MOD NRDUCELLCHNCOVALGO: NrDuCellId=0, UICoverageAlgoSwitch=TURBO\_UPLINK\_SW-0;
- **Determine whether to restore the settings of certain parameters by performing the following operations based on network conditions:**  
//Disabling uplink scheduling enhancement  
MOD NRDUCELLPUSCH: NrDuCellId=0, UIPuschAlgoSwitch=UL\_SCHEDULE\_ENH\_SW-0;  
//Disabling uplink interference randomization-based scheduling enhancement  
MOD NRDUCELLPUSCH: NrDuCellId=0,  
UIPuschAlgoSwitch=UL\_INTRF\_RANDOMIZED\_SCH\_ENH\_SW-0;  
//Disabling optimal paired RB selection  
MOD NRDUCELLULMIMO: NrDuCellId=0, UIMuMimoAlgoSwitch=OPT\_PAIR\_RB\_SEL\_SW-0;  
//Disabling uplink frequency selective scheduling optimization  
MOD NRDUCELLPUSCH: NrDuCellId=0, PuschPerformanceSwitch=UL\_FREQ\_SEL\_SCH\_OPT\_SW-0;

```
//Disabling alignment pairing optimization for large-packet UEs
MOD NRDUCELLULMIMO: NrDuCellId=0, ULMuMimoAlgoSwitch=BIG_PKT_ALIGN_PAIR_OPT_SW-0;
//Disabling alignment pairing optimization for small-packet UEs
MOD NRDUCELLULMIMO: NrDuCellId=0, ULMuMimoAlgoSwitch=SML_PKT_ALIGN_PAIR_OPT_SW-0;
//Deselecting the MU_AMC_OPT_SW option
MOD NRDUCELLULAMC: NrDuCellId=0, ULAmcAlgoSw=MU_AMC_OPT_SW-0;
//Disabling average SINR optimization in uplink scheduling
MOD NRDUCELLULAMC: NrDuCellId=0, ULAmcAlgoSw=UL_SCH_AVG_SINR_OPT_SW-0;
//Disabling SR-based smart scheduling
MOD NRDUCELLPUSCH: NrDuCellId=0, ULPuschAlgoSwitch=SR_SMART_SCH_SW-0;
//Disabling uplink non-consecutive scheduling optimization
MOD NRDUCELLPUSCH: NrDuCellId=0, PuschPerformanceSwitch=UL_NONCONSEC_SCH_OPT_SW-0;
//Restoring the MuRbRatioThld parameter
MOD NRDUCELLULMIMO: NrDuCellId=0, MuRbRatioThld=0;
//Restoring the ULMuMimoSinrThld parameter
MOD NRDUCELLULMIMO: NrDuCellId=0, ULMuMimoSinrThld=-200;
//Restoring the UL256QamSwitch parameter
MOD NRDUCELLALGOSWITCH: NrDuCellId=0, UL256QamSwitch=UL_256QAM_OFF;
//Deselecting the UL_256QAM_ADAPT_POWER_OPT_SW option
MOD NRDUCELLULAMC: NrDuCellId=0, ULAmcAlgoSw=UL_256QAM_ADAPT_POWER_OPT_SW-0;
```

#### 4.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

##### NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

### 4.4.2 Activation Verification

- RRC-reconfiguration-based dynamic waveform switching
  - This function has taken effect if the uplink user-perceived rate at the coverage edge ((**N.ThpVol.UL.Edge** – **N.ThpVol.UE.UL.SmallPkt.Edge**)/**N.ThpTime.UE.UL.RmvSmallPkt.Edge**) increases after this function is enabled.
  - Check the values of **N.ChMeas.RB.PUSCH.64QAMTP.RANK1.MCS.0** to **N.ChMeas.RB.PUSCH.64QAMTP.RANK1.MCS.27** in the function subset **ChMeas.PUSCH.64QAMTP.MCS.DuCell**. If some of the values change from zeros to non-zero values after this function is enabled, this function has taken effect.
- Frequency selective scheduling under weak uplink coverage
 

After this function is enabled, check the **Uplink Resource Block Utilizing Rate (DU)** or the uplink user-perceived rate at the coverage edge. If either value increases, the function has taken effect.
- Spatial channel estimation under weak uplink coverage
 

After this function is enabled, check the uplink user-perceived rate at the coverage edge. If the value increases, the function has taken effect.
- SR-based scheduling optimization
 

After this function is enabled, check the uplink user-perceived rate at the coverage edge. If the value increases, the function has taken effect.
- Uplink SINR filtering optimization
 

After this function is enabled, if the uplink BLER decreases for moving UEs that require a long time to complete SRS frequency hopping over the entire bandwidth, this function has taken effect.

- Full-power dual transmission  
After full-power dual transmission is enabled, if the average PUSCH RSRP increases, this function has taken effect.
- Uplink  $\pi/2$ -BPSK modulation  
If the value of **N.UL.SCH.HalfPiBPSK.TB** is not 0 after this function is enabled, this function has taken effect.
- Uplink  $\pi/2$ -BPSK modulation power boost  
If *powerBoostPi2BPSK* in the *UplinkConfig* IE is set after this function is enabled, this function has taken effect.
- Uplink low-PAPR DMRS  
If the *dmrs-UplinkTransformPrecoding-r16* IE in *DMRS-UplinkConfig* is set by the base station after this function is enabled, this function has taken effect.
- Closed-loop PUSCH antenna selection
  - On the MAE-Access, choose **Monitor > Signaling Trace > Signaling Trace Management**.
  - In the navigation tree on the left of the **Signaling Trace Management** tab page, choose **NR > Application Layer > Uu Interface Trace**.
  - Expand *RRC\_RECFG*, *srs-Config*, and *srs-ResourceSetToAddModList*, and then check the *srs-ResourceSetId:0x0* IE. If this IE has the values *SRS-ResourceId:0x0* and *SRS-ResourceId:0x1*, this function has taken effect.

#### NOTE

Uplink user-perceived rate at the coverage edge =  $(N.ThpVol.UL.Edge - N.ThpVol.UE.UL.SmallPkt.Edge) / N.ThpTime.UE.UL.RmvSmallPkt.Edge$

Average PUSCH RSRP =  $(-130 \times N.UL.PUSCH.RSRP.Index0 - 127 \times N.UL.PUSCH.RSRP.Index1 - 123 \times N.UL.PUSCH.RSRP.Index2 - 117 \times N.UL.PUSCH.RSRP.Index3 - 113 \times N.UL.PUSCH.RSRP.Index4 - 107 \times N.UL.PUSCH.RSRP.Index5 - 103 \times N.UL.PUSCH.RSRP.Index6 - 97 \times N.UL.PUSCH.RSRP.Index7 - 93 \times N.UL.PUSCH.RSRP.Index8 - 87 \times N.UL.PUSCH.RSRP.Index9 - 83 \times N.UL.PUSCH.RSRP.Index10 - 80 \times N.UL.PUSCH.RSRP.Index11) / (N.UL.PUSCH.RSRP.Index0 + N.UL.PUSCH.RSRP.Index1 + N.UL.PUSCH.RSRP.Index2 + N.UL.PUSCH.RSRP.Index3 + N.UL.PUSCH.RSRP.Index4 + N.UL.PUSCH.RSRP.Index5 + N.UL.PUSCH.RSRP.Index6 + N.UL.PUSCH.RSRP.Index7 + N.UL.PUSCH.RSRP.Index8 + N.UL.PUSCH.RSRP.Index9 + N.UL.PUSCH.RSRP.Index10 + N.UL.PUSCH.RSRP.Index11)$

## 4.4.3 Network Monitoring

After this feature is enabled, monitor its running status by checking the increase in the uplink user-perceived rate at the coverage edge.

#### NOTE

Uplink user-perceived rate at the coverage edge =  $(N.ThpVol.UL.Edge - N.ThpVol.UE.UL.SmallPkt.Edge) / N.ThpTime.UE.UL.RmvSmallPkt.Edge$



# 5 Turbo Uplink 2.0

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## 5.1 Principles

### 5.1.1 Interference Randomization Under Weak Uplink Coverage

As the transmit power of UEs is limited, the uplink user-perceived rate may be affected when the UEs move to weak coverage areas. In addition, the start positions of RB allocation are the same for cells with the same result of PCI mod 3, and there is a high probability that allocated RBs overlap. This leads to strong uplink interference between these cells, affecting the uplink user-perceived rate.

To address these issues, interference randomization under weak uplink coverage is introduced. After this function is enabled, **the base station utilizes the frequency selective characteristics of channels (that is, different SINRs in different frequency-domain positions). The base station schedules UEs under weak coverage in RB positions with high SINRs**, as shown in [Figure 5-1](#). Moreover, **this function staggers the start positions of RBs for cells with the same PCI-mod-3 value, so that allocation of RBs for uplink scheduling starts from different positions**, as shown in [Figure 5-2](#). This reduces the inter-cell uplink interference and increases the average uplink UE throughput.

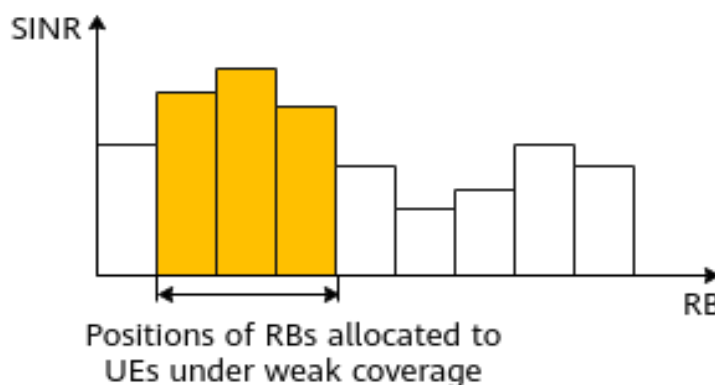
This function is controlled by the **UL\_WEAK\_COV\_INTRF\_RAND\_SW** option of the **NRDUCellChnCovAlgo.UlCoverageAlgoSwitch** parameter.

As cell-edge UEs are more sensitive to interference, beam selection is likely to be inaccurate when uplink SRS interference fluctuates greatly, affecting the uplink user-perceived rate. Therefore, it is recommended that the **NRDUCellSrsMeas.SrsRsrpExcThld** parameter be set to **DB6** to reduce the impact of uplink SRS interference fluctuation on uplink user experience.

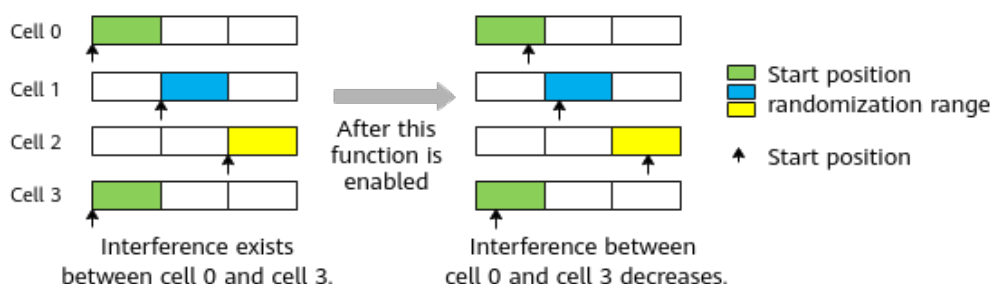
After interference randomization under weak uplink coverage is enabled, the RRC setup success rate may fluctuate. To address this issue, allocation adjustment for uplink retransmission is introduced. After this function is enabled, randomization-based staggering is not performed on the start positions of retransmission resources during initial access of a UE. This increases the RRC setup success rate. This function is controlled by the **UL\_RETRANS\_ALLOC\_ADJ\_SW** option of the

**NRDUCellChnCovAlgo.UlCoverageAlgoSwitch** parameter. For cells where interference randomization under weak uplink coverage is to be enabled, it is recommended that allocation adjustment for uplink retransmission also be enabled.

**Figure 5-1** Positions of RBs allocated to UEs under weak coverage



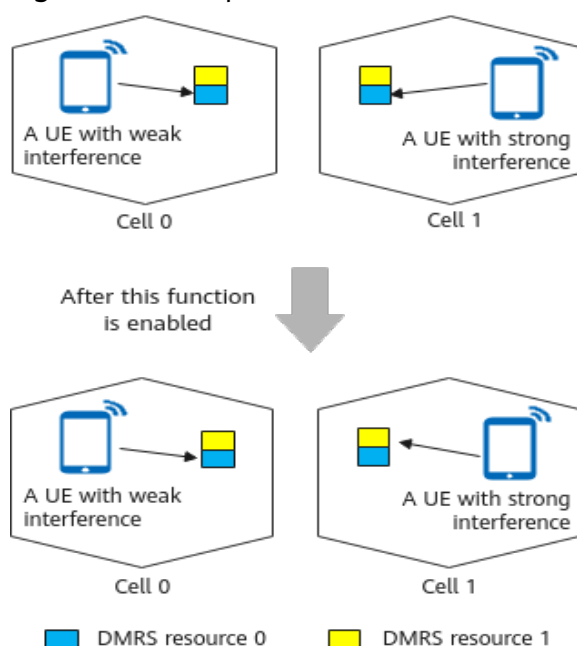
**Figure 5-2** Start positions of RBs in adjacent cells



## 5.1.2 DMRS Interference Reduction

As the transmit power of UEs is limited, the capability of UEs to resist interference decreases when they move to weak coverage areas. To address this issue, DMRS interference reduction is introduced. With this function, the base station adaptively distinguishes UEs with strong interference from UEs with weak interference based on UE locations. (UEs with strong interference refer to UEs that cause strong interference to other UEs, and UEs with weak interference refer to UEs that cause weak interference to other UEs.) **The base station allocates different DMRS resources to different types of UEs, as shown in Figure 5-3. This reduces the DMRS interference caused by UEs with strong interference to UEs with weak interference, improving the uplink user-perceived rate of UEs under weak coverage.** This function is controlled by the **DMRS\_RESOURCE\_ADJ\_SW** option of the **NRDUCellChnCovAlgo.UlCoverageAlgoSwitch** parameter.

**Figure 5-3 Principle of DMRS interference reduction**

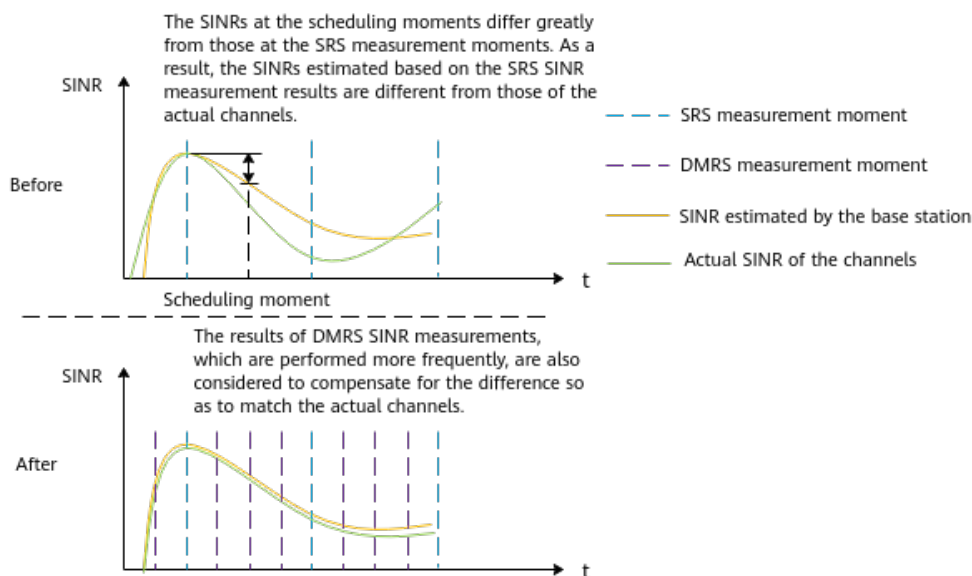


### 5.1.3 High-Reliability Uplink MCS Selection

The SINR of an uplink channel reflects the quality of this channel. If the channel changes rapidly, the SINR at the gNodeB measurement moment differs greatly from that at the scheduling moment. If the measured SINR cannot match rapid channel changes in a timely manner, the selected uplink MCS is inaccurate. If the selected uplink MCS is too high, the uplink BLER increases and the uplink UE rate decreases. If the selected uplink MCS is too low, the uplink UE rate is limited because the channel capability is not fully utilized.

To address this issue, high-reliability uplink MCS selection is introduced. **Figure 5-4** shows how this function works. After this function is enabled, **the gNodeB measures both SRS SINRs and DMRS SINRs. Based on the measurement results, it adaptively adjusts the scheduling-moment SINRs to match the actual channels. This prevents the selected MCS from being too high or too low, which improves the reliability of the uplink MCS selection and increases the uplink user-perceived rate.** This function is controlled by the **UL\_HIGH\_RELIAB\_MCS\_SEL\_SW** option of the **NRDUCellULAmc.UlAmcPerformanceSw** parameter.

**Figure 5-4 High-reliability uplink MCS selection**



If the SRS measurement interval is long and the UE traffic volume is low, the measured SINRs cannot match channel changes in a timely manner even after high-reliability uplink MCS selection is enabled. This is because the numbers of SRS and DMRS measurements performed are less than required. To address this issue, the filtering coefficients for uplink DMRS SINRs can be adjusted to enable the SINRs to quickly match channel changes. The adjustment can be enabled by selecting the **UL\_DMRS\_FILTER\_COEFF\_ADAPT\_SW** option of the **NRDUCellUIMc.UIMcPerformanceSw** parameter. After this option is selected, the gNodeB adaptively determines the filtering coefficients for uplink DMRS SINRs so that the filtered SINRs can better match the actual channels.

## 5.1.4 Uplink Data Compression

Uplink data compression is a trial function. Trial functions are functions that are not yet ready for full commercial release for certain reasons. For example, the industry chain (terminals/CN) may not be sufficiently compatible. However, these functions can still be used for testing purposes or commercial network trials. Anyone who desires to use the trial functions shall contact Huawei and enter into a memorandum of understanding (MoU) with Huawei prior to an official application of such trial functions. Trial functions are not for sale in the current version but customers may try them for free.

Customers acknowledge and undertake that trial functions may have a certain degree of risk due to absence of commercial testing. Before using them, customers shall fully understand not only the expected benefits of such trial functions but also the possible impact they may exert on the network. In addition, customers acknowledge and undertake that since trial functions are free, Huawei is not liable for any trial function malfunctions or any losses incurred by using the trial functions. Huawei does not promise that problems with trial functions will be resolved in the current version. Huawei reserves the rights to convert trial functions into commercial functions in later R/C versions. If trial functions are converted into commercial functions in a later version, customers shall pay a licensing fee to obtain the relevant licenses prior to using the said commercial

functions. If a customer fails to purchase such a license, the trial function(s) will be invalidated automatically when the product is upgraded.

Principles of Uplink Data Compression

Uplink data compression (UDC) uses the 3GPP Deflate compression algorithm to compress uplink data packets. (For details, see section 5.14.1 "UDC protocol" in 3GPP TS 38.323 V17.5.0.) This reduces resource consumption over the air interface and improves uplink data transmission capability. This function applies to the PDCP layers of the UE and gNodeB. After this function is enabled, the UE compresses uplink data packets that are not compressed or encrypted and then the gNodeB decompresses the compressed packets to reduce the amount of data transmitted over the air interface. This function is illustrated in Figure 5-5 and its principles are described as follows:

1.

The UE and gNodeB separately maintain buffers with the same size and stored content as a reference to compare with raw data. The greater the repetition of information in the buffer within the raw data, the better the compression effect.
2.

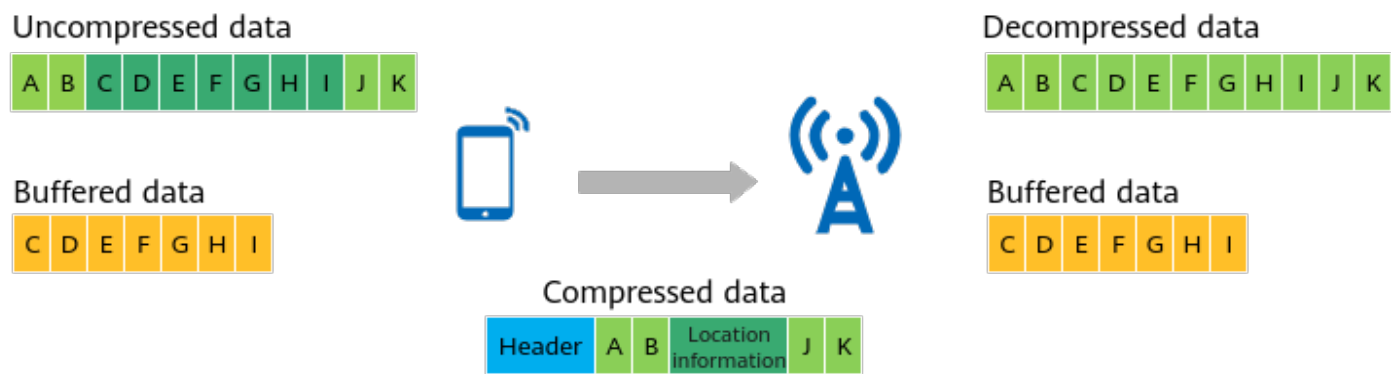
Compression on the UE side
  - If the to-be-transmitted data includes content that is the same as the data in the buffer, the content is not transmitted; instead, the location information of the content in the buffer is transmitted.
  - If the to-be-transmitted data has content different from the data in the buffer, the content is directly transmitted.

A UDC packet header is used to indicate whether data is compressed. The final compressed data consists of the UDC packet header and the content that is different from the buffered data.
3.

Decompression on the gNodeB side

After receiving the data, the gNodeB determines whether the data has been compressed based on the packet header. If the data has been compressed, the gNodeB reads the duplicate content from the buffer based on the location information and reassembles the duplicate content with the non-duplicate content to restore the original data.

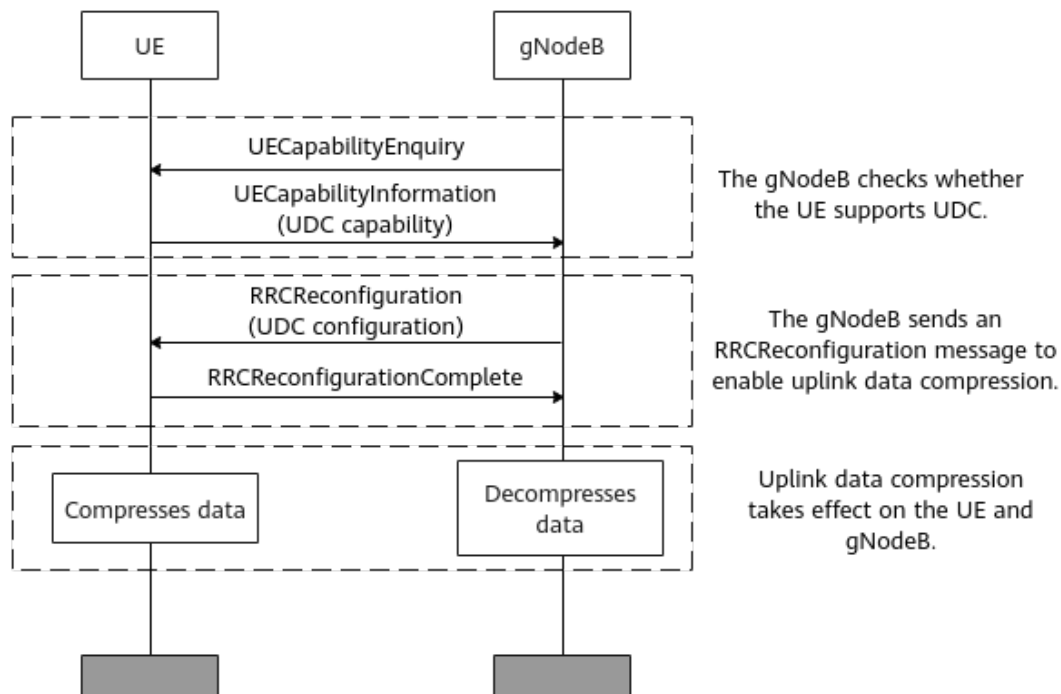
Figure 5-5 Principles of uplink data compression



## Uplink Data Compression Procedure

Figure 5-6 shows the uplink data compression procedure.

Figure 5-6 Uplink data compression procedure



1. The gNodeB checks whether the UE supports UDC.
  - a. The gNodeB sends a **UECapabilityEnquiry** message to the UE.
  - b. The UE responds to the gNodeB with a **UECapabilityInformation** message. If the **udc-r17** field is included in the message, the UE supports UDC.
2. The gNodeB sends an **RRCReconfiguration** message to enable uplink data compression.
  - a. The gNodeB sends an **RRCReconfiguration** message to the UE, instructing the UE to enable uplink data compression.
  - b. The UE responds to the gNodeB with an **RRCReconfigurationComplete** message.
3. Uplink data compression takes effect on the UE and gNodeB.  
The UE compresses uplink data packets, and the gNodeB decompresses the compressed packets.

## Parameters Related to Uplink Data Compression

Uplink data compression can be configured at the cell level or QCI level.

- Cell-level parameter configuration
  - **UDC\_SW** option of the **NRDUCellChnCovAlgo.UlCoverageAlgoSwitch** parameter: If this option is selected, uplink data compression can take effect for all uplink data packets transmitted in a cell for data services.

- **UPLINK\_SIG\_COMPRESS\_SW** option of the **NRDUCellChnCovAlgo.UlCoverageAlgoSwitch** parameter: If this option is selected, uplink data compression can take effect for all uplink data packets transmitted in a cell for voice services.
- **NRCCellServExp.UdcBufferSize**: This parameter specifies the buffer size on the gNodeB side. The actual buffer size is equal to the value of this parameter or the buffer size reported by the UE, whichever is smaller. A smaller buffer size results in a lower compression efficiency but a shorter decompression delay. A larger buffer size produces the opposite effects.
- QCI-level parameter configuration
  - QCI\_UDC\_SW** option of the **gNBQciBearer.PdcpAlgoSwitch** parameter: If this option is selected, uplink data compression can take effect for bearers with the specified QCI.
    - For such bearers carrying data services, if the **UDC\_SW** option of the cell-level parameter **NRDUCellChnCovAlgo.UlCoverageAlgoSwitch** is selected, this cell-level parameter setting takes effect; otherwise, the QCI-level parameter setting takes effect.
    - For such bearers carrying voice services (with QCI 5 or 69): If the **UPLINK\_SIG\_COMPRESS\_SW** option of the cell-level parameter **NRDUCellChnCovAlgo.UlCoverageAlgoSwitch** is selected, this cell-level parameter setting takes effect; otherwise, the QCI-level parameter setting takes effect.

A maximum of two uplink DRBs can be compressed for a single UE. This function is supported only in SA networking.

## Uplink Data Compression Based on the SIP Standard Dictionary

In the initial phase of voice call setup, the initial buffers of the UE and gNodeB are empty. Therefore, the Session Initiation Protocol (SIP) signaling compression efficiency is relatively low, which affects the voice call delay.

For bearers of 5QI 5 and 5QI 69 (mainly involving SIP signaling), 3GPP has defined a preset standard dictionary for SIP and SDP, which refers to buffered data predefined on the UE and gNodeB sides. Accordingly, uplink data compression based on the SIP standard dictionary is introduced, which uses the standard dictionary for SIP and SDP. In the initial phase of bearer setup, the UE and gNodeB fill in the buffers with contents in the standard dictionary as a basis for data compression. This function effectively solves the problem that the initial buffers are empty at the initial stage of bearer setup and implements efficient data compression, thereby reducing the voice call delay of cell-edge UEs.

This function works as follows:

1. The UE reports its UDC capability. The UDC standard dictionary capability is indicated by the **standardDictionary-r17** field in **udc-r17**.
2. The UE and the gNodeB perform the following operations after completing the setup of an RRC connection:
  - a. The gNodeB sends an RRCReconfiguration message to the UE, instructing the UE to enable uplink data compression based on the SIP standard dictionary.
  - b. The UE responds to the gNodeB with an RRCReconfigurationComplete message.

3. The UE compresses uplink SIP signaling packets based on the preset UDC standard dictionary, and the gNodeB decompresses the compressed packets.

This function is controlled by the **UL\_SIP\_UDC\_DICT\_SW** option of the **NRCellServExp.VoiceAlgoSwitch** parameter. Before enabling this function, select the **UPLINK\_SIG\_COMPRESS\_SW** option of the **NRDUCellChnCovAlgo.UlCoverageAlgoSwitch** parameter or the **QCI\_UDC\_SW** option of the **gNBQciBearer.PdcpAlgoSwitch** parameter for the corresponding QCI.

This function applies only to SA networking. Bearers of 5QI 5 and 5QI 69 are used to carry SIP signaling for VoNR services. Therefore, this function takes effect only for such bearers.

### 5.1.5 Probability-based Uplink MCS Selection

Due to the impact of channel variations on signals, the SINR during gNodeB measurement varies from that during scheduling. In this background, probability-based uplink MCS selection is introduced. **Based on the probability of different SINRs used at historical measurement and scheduling moments, the gNodeB estimates the channel quality fluctuation level and SINR adjustment amount for UEs so that the sum of SINR during measurement and the SINR adjustment amount can match the actual channel quality fluctuation as much as possible.** This will make uplink MCS selection more reliable for UEs and increase their uplink user-perceived rates. This function is controlled by the **UL\_PBMCSSEL\_SW** option of the **NRDUCellUIMc.UlAmcPerformanceSw** parameter.

To further improve the accuracy of estimating the channel quality fluctuation level, the **UL\_HIGH\_RELIAB\_MCS\_SEL\_SW** option of the **NRDUCellUIMc.UlAmcPerformanceSw** parameter can be selected when this function is enabled.

To reduce the impact of SINR adjustment on RB allocation and improve the reliability of uplink MCS selection in frequency selective scheduling scenarios, the **MCS\_TABLE\_BASED\_SCH\_RB\_ADJ\_SW** option of the **NRDUCellUISchRes.UlSchResAlgoSwitch** parameter can also be selected.

### 5.1.6 Time-Domain Adaptive Filtering

In addition to channel environment changes, the time-varying nature of interference signals also affects the accuracy of channel estimation. To further improve channel estimation accuracy, time-domain adaptive filtering is introduced. In massive MIMO cells with multiple RX channels, **the base station adaptively adjusts time-domain filtering coefficients based on the PDP of PUSCH DMRs and the time-domain changes in the strength of useful signals and interference signals on each RX channel. This increases the SINR of received signals, improving the accuracy of channel estimation.** This function is controlled by the **PUSCH\_TIME\_DOM\_ADAPT\_FILTER\_SW** option of the **NRDUCellChnCovAlgo.UlCoverageAlgoSwitch** parameter. This function does not take effect for MU-MIMO UEs or UEs running URLLC services.

## 5.2 Network Analysis



## 5.2.1 Benefits

To maximize the gains of the Turbo Uplink 2.0 feature, it is recommended that all the following conditions be met before this feature is enabled:

- There are UEs in weak coverage areas. Specifically, the average number of active UEs at the uplink coverage edges in a cell (**N.User.RRCCConn.Active.UL.Edge.Avg**) is greater than or equal to 0.5, and the proportion of such UEs (**N.User.RRCCConn.Active.UL.Edge.Avg** / **N.User.RRCCConn.Active.UL.Avg** x 100%) is greater than 5%.
- There is no remote interference (that is, **N.GAP.LastSymbol.Pwr** – **N.UL.Last.Symbol13.Pwr** ≤ 5 dB), and the average uplink interference (**N.UL.NI.Avg**) is between –110 dBm and –95 dBm.
- The uplink PRB usage (**Uplink Resource Block Utilizing Rate (DU)**) is less than 40%.
- The average SRS interference (**N.SRS.NI.Avg**) is greater than or equal to –110 dBm.
- The proportion of RBs that are not preallocated ((**N.PR.B.PUSCH.Used.Avg** – **N.PR.B.UL.Prealloc.Used.Avg**) / **N.PR.B.UL.PUSCH.Avail.Avg** x 100%) is greater than 10%.
- The uplink initial block error rate (IBLER) of the cell is greater than 8%.

Turbo Uplink 2.0 increases the uplink user-perceived rate at the coverage edge and the **User Uplink Average Throughput (DU)**. The benefits of each function are as follows:

- Interference randomization under weak uplink coverage
  - If the uplink PRB usage of a cell is less than or equal to 30%, this function increases the uplink user-perceived rate at the coverage edge. The specific benefit depends on the frequency selective characteristics and interference. The more significant the characteristics and the stronger the interference, the greater the benefit. In addition, this function reduces the average uplink interference in the cell (**N.UL.NI.Avg**).
  - If the uplink PRB usage of a cell is greater than 30%, there is no negative impact.
  - When interference randomization under weak uplink coverage is enabled, allocation adjustment for uplink retransmission may increase the RRC setup success rate.
- DMRS interference reduction
  - If the uplink PRB usage of a cell is less than or equal to 30%, this function increases the uplink user-perceived rate at the coverage edge. The benefit depends on uplink interference. The stronger the interference, the greater the benefit.
  - If the uplink PRB usage of a cell is greater than 30%, there is no negative impact.
- High-reliability uplink MCS selection
  - In scenarios where channel conditions fluctuate significantly and the uplink IBLER is high, this function decreases the uplink IBLER and uplink retransmission proportion and increases the uplink user-perceived rate at the coverage edge and the **User Uplink Average Throughput (DU)**.

- To avoid negative impacts of this function in CCE insufficiency scenarios (where the uplink and downlink CCE allocation success rates are less than 95%), it is recommended that the **PUSCH\_DTX\_AGG\_LVL\_ADAPT\_SW** option of the **NRDuCellPdcch.PdcchAlgoEnhSwitch** parameter be deselected before function activation to preferentially resolve CCE insufficiency.
- Uplink data compression
  - This function improves the compression efficiency of uplink data packets, decreases the amount of data transmitted over the air interface, and reduces the transmission delay for cell-edge UEs. The higher the compression efficiency of uplink data packets, the greater the decrease in the amount of data transmitted over the air interface.
  - After this function takes effect, compressed data packets carry compression indication information, causing extra overhead. Therefore, if the repetition rate of uplink data is not high (for example, uplink data packets have been compressed or encrypted), the size of data packets after compression may be greater than that before compression, leading to possible negative impacts.
  - After this function takes effect, a larger buffer size results in a higher compression efficiency but a longer decompression delay. A smaller buffer size produces the opposite effects.
  - Uplink data compression based on the SIP standard dictionary improves the data compression efficiency in the initial phase of uplink SIP signaling transmission, thereby reducing the voice call setup delay for cell-edge UEs.
- Probability-based uplink MCS selection

This function increases the uplink user-perceived rate at the coverage edge. The gains depend on the proportion of cell-edge UEs in weak coverage areas. The larger the proportion, the greater the gains. In addition, the **User Uplink Average Throughput (DU)** and **Cell Uplink Average Throughput (DU)** increase or remain unchanged. The increases depend on the traffic model and UE distribution.
- Time-domain adaptive filtering

After this function takes effect, the average uplink MCS index of the cell slightly increases or remains unchanged, the uplink BLER slightly decreases or remains unchanged, and the uplink user-perceived rate at the coverage edge increases. In addition, the **User Uplink Average Throughput (DU)** and **Cell Uplink Average Throughput (DU)** slightly increase or remain unchanged.

#### NOTE

- Uplink user-perceived rate at the coverage edge =  $(N.ThpVol.UL.Edge - N.ThpVol.UE.UL.SmallPkt.Edge) / N.ThpTime.UE.UL.RmvSmallPkt.Edge$
- Compression efficiency =  $(1 - N.PDCP.Vol.UL.UDC.BefDecomp / N.PDCP.Vol.UL.UDC.AftDecomp) \times 100\%$
- IBLER of uplink scheduling using QPSK and  $\pi/2$ -BPSK =  $(N.UL.SCH.HalfPiBPSK.ErrTB.Ibler + N.UL.SCH.QPSK.ErrTB.Ibler) / (N.UL.SCH.HalfPiBPSK.TB + N.UL.SCH.QPSK.TB) \times 100\%$
- Uplink IBLER of a cell =  $(N.UL.SCH.HalfPiBPSK.ErrTB.Ibler + N.UL.SCH.QPSK.ErrTB.Ibler + N.UL.SCH.16QAM.ErrTB.Ibler + N.UL.SCH.64QAM.ErrTB.Ibler + N.UL.SCH.256QAM.ErrTB.Ibler) / (N.UL.SCH.HalfPiBPSK.TB + N.UL.SCH.QPSK.TB + N.UL.SCH.16QAM.TB + N.UL.SCH.64QAM.TB + N.UL.SCH.256QAM.TB) \times 100\%$
- Proportion of cell-edge UEs in weak coverage areas =  $N.User.RRConn.Active.UL.Edge.Avg / N.User.RRConn.Active.UL.Avg \times 100\%$

## 5.2.2 Impacts

The impacts between this function and other functions are described in [5.3.2.3 Function Impacts](#).

- Interference randomization under weak uplink coverage  
After interference randomization under weak uplink coverage takes effect, the RRC setup success rate may fluctuate. For cells where interference randomization under weak uplink coverage is to be enabled, it is recommended that allocation adjustment for uplink retransmission also be enabled.
- DMRS interference reduction  
None
- High-reliability uplink MCS selection
  - After this function takes effect, the values of the following indicators fluctuate or remain unchanged, depending on the traffic model and the proportions of UEs in the cell center, at a medium distance from the cell center, and at the cell edge.
    - Uplink PRB usage (**Uplink Resource Block Utilizing Rate (DU)**)
    - Average uplink MCS index of a cell
    - Average uplink rank in a cell
    - Uplink RBLER of a cell
  - After this function takes effect, the average gNodeB board user-plane CPU usage (**VS.NRBoard.UPlane.CPULoad.Avg**) slightly increases.
  - After this function takes effect, the proportion of UEs with high uplink throughput (which is greater than 30 Mbit/s) slightly decreases.  
Proportion of UEs with high uplink throughput (which is greater than 30 Mbit/s) =  $SUM(N.Thp.UL.Samp.Indexk) / SUM(N.Thp.UL.Samp.Indexm)$ , where  $k$  is in the range of 8 to 14 and  $m$  is in the range of 0 to 14.
- Uplink data compression  
After this function takes effect, the average gNodeB board user-plane CPU usage (**VS.NRBoard.UPlane.CPULoad.Avg**) increases, the uplink PRB usage

(**Uplink Resource Block Utilizing Rate (DU)**) decreases or remains unchanged, the uplink MCS index fluctuates, and the uplink user-perceived rate at the coverage edge fluctuates.

- Probability-based uplink MCS selection
  - After this function takes effect, the values of the following indicators fluctuate or remain unchanged, depending on the proportions of UEs in the cell center, at a medium distance from the cell center, and at the cell edge:
    - Uplink PRB usage (**Uplink Resource Block Utilizing Rate (DU)**)
    - Cell uplink IBLER. When there are a large number of cell-center UEs, the uplink IBLER decreases in the cell. When there are a large number of cell-edge UEs, the uplink IBLER increases in the cell.
    - Average uplink MCS index
    - CCE usage =  $\text{N.CCE.Used.Avg} / \text{N.CCE.Avail.Avg} \times 100\%$
    - Contention-based access success rate and non-contention-based access success rate
  - After this function takes effect, the average gNodeB board user-plane CPU usage (**VS.NRBoard.UPlane.CPULoad.Avg**) slightly increases.
  - After this function takes effect, the proportion of UEs with high uplink throughput (which is greater than 30 Mbit/s) slightly decreases. Proportion of UEs with high uplink throughput (which is greater than 30 Mbit/s) =  $\text{SUM}(\text{N.Thp.UL.Samp.Index}k) / \text{SUM}(\text{N.Thp.UL.Samp.Index}m)$ , where  $k$  is in the range of 8 to 14 and  $m$  is in the range of 0 to 14.
- Time-domain adaptive filtering  
None

#### NOTE

- Average uplink rank in a cell =  $(\text{N.PUSCH.TbUL.Rank1} \times 1 + \text{N.PUSCH.TbUL.Rank2} \times 2) / (\text{N.PUSCH.TbUL.Rank1} + \text{N.PUSCH.TbUL.Rank2})$
- Uplink RBLER of a cell =  $(\text{N.UL.SCH.HalfPiBPSK.ErrTB.Rbler} + \text{N.UL.SCH.QPSK.ErrTB.Rbler} + \text{N.UL.SCH.16QAM.ErrTB.Rbler} + \text{N.UL.SCH.64QAM.ErrTB.Rbler} + \text{N.UL.SCH.256QAM.ErrTB.Rbler}) / (\text{N.UL.SCH.HalfPiBPSK.TB} + \text{N.UL.SCH.QPSK.TB} + \text{N.UL.SCH.16QAM.TB} + \text{N.UL.SCH.64QAM.TB} + \text{N.UL.SCH.256QAM.TB}) \times 100\%$
- Average uplink MCS index in a cell =  $(0 \times \text{N.ChMeas.PUSCH.MCS.0} + 1 \times \text{N.ChMeas.PUSCH.MCS.1} + 2 \times \text{N.ChMeas.PUSCH.MCS.2} + \dots + 28 \times \text{N.ChMeas.PUSCH.MCS.28}) / (\text{N.ChMeas.PUSCH.MCS.0} + \text{N.ChMeas.PUSCH.MCS.1} + \text{N.ChMeas.PUSCH.MCS.2} + \dots + \text{N.ChMeas.PUSCH.MCS.28})$
- Uplink IBLER of a cell =  $(\text{N.UL.SCH.HalfPiBPSK.ErrTB.Ibler} + \text{N.UL.SCH.QPSK.ErrTB.Ibler} + \text{N.UL.SCH.16QAM.ErrTB.Ibler} + \text{N.UL.SCH.64QAM.ErrTB.Ibler} + \text{N.UL.SCH.256QAM.ErrTB.Ibler}) / (\text{N.UL.SCH.HalfPiBPSK.TB} + \text{N.UL.SCH.QPSK.TB} + \text{N.UL.SCH.16QAM.TB} + \text{N.UL.SCH.64QAM.TB} + \text{N.UL.SCH.256QAM.TB}) \times 100\%$
- Contention-based access success rate =  $\text{N.RA.Contention.Resolution.Succ} / \text{N.RA.Contention.Att} \times 100\%$
- Non-contention-based access success rate =  $\text{N.RA.Dedicated.Msg3} / \text{N.RA.Dedicated.Att} \times 100\%$

## 5.3 Requirements

### 5.3.1 Licenses

| Feature ID  | Feature Name     | Model            | NE     | Sales Unit |
|-------------|------------------|------------------|--------|------------|
| FOFD-081202 | Turbo Uplink 2.0 | NR0S00TBU<br>L20 | gNodeB | Per Cell   |

#### NOTE

The insufficiency of certain feature licenses affects cell activation. For details, see the **Is Cell Activation Affected by License Insufficiency** column in *License Control Item Lists*.

### 5.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

#### 5.3.2.1 Prerequisite Functions

- Interference randomization under weak uplink coverage

| Function Name              | Function Switch                                                                                      | Reference                               | Description                                                                                                                |
|----------------------------|------------------------------------------------------------------------------------------------------|-----------------------------------------|----------------------------------------------------------------------------------------------------------------------------|
| Turbo Uplink Switch        | <b>TURBO_UPLINK_SW</b> option of the <b>NRDUCellChnCoVAlgo.UICoverageAlgoSwitch</b> parameter        | <i>Turbo Uplink (Low-Frequency TDD)</i> | The Turbo Uplink Phase2 Switch can be turned on only if the Turbo Uplink Switch is turned on.                              |
| Turbo Uplink Phase2 Switch | <b>TURBO_UPLINK_PHASE2_SW</b> option of the <b>NRDUCellChnCoVAlgo.UICoverageAlgoSwitch</b> parameter | <i>Turbo Uplink (Low-Frequency TDD)</i> | Interference randomization under weak uplink coverage can take effect only if the Turbo Uplink Phase2 Switch is turned on. |

| Function Name                                         | Function Switch                                                                                         | Reference                               | Description                                                                                                                               |
|-------------------------------------------------------|---------------------------------------------------------------------------------------------------------|-----------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------|
| Interference randomization under weak uplink coverage | <b>UL_WEAK_COV_INTRF_RAND_SW</b> option of the <b>NRDUCellChnCoVAlgo.UICoverageAlgoSwitch</b> parameter | <i>Turbo Uplink (Low-Frequency TDD)</i> | Allocation adjustment for uplink retransmission can take effect only if interference randomization under weak uplink coverage is enabled. |

- DMRS interference reduction/Probability-based uplink MCS selection/Time-domain adaptive filtering

| Function Name              | Function Switch                                                                                      | Reference                               | Description                                                                                                                                                                                           |
|----------------------------|------------------------------------------------------------------------------------------------------|-----------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Turbo Uplink Switch        | <b>TURBO_UPLINK_SW</b> option of the <b>NRDUCellChnCoVAlgo.UICoverageAlgoSwitch</b> parameter        | <i>Turbo Uplink (Low-Frequency TDD)</i> | The Turbo Uplink Phase2 Switch can be turned on only if the Turbo Uplink Switch is turned on.                                                                                                         |
| Turbo Uplink Phase2 Switch | <b>TURBO_UPLINK_PHASE2_SW</b> option of the <b>NRDUCellChnCoVAlgo.UICoverageAlgoSwitch</b> parameter | <i>Turbo Uplink (Low-Frequency TDD)</i> | The following functions can take effect only if the Turbo Uplink Phase2 Switch is turned on: DMRS interference reduction, probability-based uplink MCS selection, and time-domain adaptive filtering. |

- High-reliability uplink MCS selection

| Function Name                         | Function Switch                                                                                      | Reference                               | Description                                                                                                           |
|---------------------------------------|------------------------------------------------------------------------------------------------------|-----------------------------------------|-----------------------------------------------------------------------------------------------------------------------|
| Turbo Uplink Phase2 Switch            | <b>TURBO_UPLINK_PHASE2_SW</b> option of the <b>NRDUCellChnCoVAlgo.UICoverageAlgoSwitch</b> parameter | <i>Turbo Uplink (Low-Frequency TDD)</i> | High-reliability uplink MCS selection can take effect only if the Turbo Uplink Phase2 Switch is turned on.            |
| Turbo Uplink Switch                   | <b>TURBO_UPLINK_SW</b> option of the <b>NRDUCellChnCoVAlgo.UICoverageAlgoSwitch</b> parameter        | <i>Turbo Uplink (Low-Frequency TDD)</i> | The Turbo Uplink Phase2 Switch can be turned on only if the Turbo Uplink Switch is turned on.                         |
| High-reliability uplink MCS selection | <b>UL_HIGH_RELIAB_MCS_SEL_SW</b> option of the <b>NRDUCellULAmc.UlAmcPerformanceSw</b> parameter     | <i>Turbo Uplink (Low-Frequency TDD)</i> | Uplink DMRS filtering coefficient adaptation can be enabled only if high-reliability uplink MCS selection is enabled. |

- Uplink data compression

| Function Name              | Function Switch                                                                                      | Reference                               | Description                                                                                   |
|----------------------------|------------------------------------------------------------------------------------------------------|-----------------------------------------|-----------------------------------------------------------------------------------------------|
| Turbo Uplink Switch        | <b>TURBO_UPLINK_SW</b> option of the <b>NRDUCellChnCoVAlgo.UICoverageAlgoSwitch</b> parameter        | <i>Turbo Uplink (Low-Frequency TDD)</i> | The Turbo Uplink Phase2 Switch can be turned on only if the Turbo Uplink Switch is turned on. |
| Turbo Uplink Phase2 Switch | <b>TURBO_UPLINK_PHASE2_SW</b> option of the <b>NRDUCellChnCoVAlgo.UICoverageAlgoSwitch</b> parameter | <i>Turbo Uplink (Low-Frequency TDD)</i> | Uplink data compression can take effect only if the Turbo Uplink Phase2 Switch is turned on.  |

| Function Name           | Function Switch                                                                                                                                                                         | Reference                               | Description                                                                                                                                                                                                                                                                                    |
|-------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Uplink data compression | <b>UPLINK_SIG_COMPRESS_SW</b> option of the <b>NRDUCellChnConfigAlgo.UlCoverageAlgoSwitch</b> parameter or <b>QCI_UDC_SW</b> option of the <b>gNBQciBearer.PdcpAlgoSwitch</b> parameter | <i>Turbo Uplink (Low-Frequency TDD)</i> | Uplink data compression based on the SIP standard dictionary takes effect only if the <b>UPLINK_SIG_COMPRESS_SW</b> option of the <b>NRDUCellChnConfigAlgo.UlCoverageAlgoSwitch</b> parameter or the <b>QCI_UDC_SW</b> option of the <b>gNBQciBearer.PdcpAlgoSwitch</b> parameter is selected. |

### 5.3.2.2 Mutually Exclusive Functions

- Interference randomization under weak uplink coverage  
None
- DMRS interference reduction  
None
- High-reliability uplink MCS selection

| Function Name                          | Function Switch                                                                                  | Reference                  | Description                                                                                                     |
|----------------------------------------|--------------------------------------------------------------------------------------------------|----------------------------|-----------------------------------------------------------------------------------------------------------------|
| High-speed Railway Superior Experience | <b>NRDUCell.HighSpeedFlag</b> set to <b>HIGH_SPEED</b>                                           | <i>High Speed Mobility</i> | When High-speed Railway Superior Experience is enabled, high-reliability uplink MCS selection must be disabled. |
| Uplink dual filtering                  | <b>UL_SRS_DMRS_DUAL_FILTER_SW</b> option of the <b>NRDUCellPusch.UlPuschAlgoSwitch</b> parameter | <i>Uplink Scheduling</i>   | High-reliability uplink MCS selection can be enabled only when uplink dual filtering is disabled.               |



- Uplink data compression

| Function Name        | Function Switch         | Reference             | Description                                                                                                                                                                                     |
|----------------------|-------------------------|-----------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| QoS Class Identifier | <b>gNBQciBearer.Qci</b> | <i>QoS Management</i> | When the <b>gNBQciBearer.Qci</b> parameter is set to <b>1</b> , <b>65</b> , or <b>66</b> , the <b>QCI_UDC_SW</b> option of the <b>gNBQciBearer.PdcpAlgoSwitch</b> parameter cannot be selected. |

- Probability-based uplink MCS selection

None

- Time-domain adaptive filtering

| Function Name                          | Function Switch                                                                              | Reference                                                | Description                                                                                       |
|----------------------------------------|----------------------------------------------------------------------------------------------|----------------------------------------------------------|---------------------------------------------------------------------------------------------------|
| High-speed Railway Superior Experience | <b>NRDUCell.HighSpeedFlag</b> set to <b>HIGH_SPEED</b>                                       | <i>High Speed Mobility</i>                               | High-speed Railway Superior Experience and time-domain adaptive filtering cannot be both enabled. |
| Distributed massive MIMO               | <b>DM_MIMO_SERVICE_SWITCH</b> option of the <b>NRDUCellAlgoSwitch.DmMimoSwitch</b> parameter | <i>Distributed Antenna Solutions (Low-Frequency TDD)</i> | Distributed massive MIMO and time-domain adaptive filtering cannot be both enabled.               |

### 5.3.2.3 Function Impacts

- Interference randomization under weak uplink coverage

| Function Name                  | Function Switch                                                                          | Reference                                               | Description                                                                                                                                                                                                               |
|--------------------------------|------------------------------------------------------------------------------------------|---------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Uplink MU grouping and pairing | <b>UL_MU_GRP_PAIR_SW</b> option of the <b>NRDUCellUIMimo.UlMuMimoAgoSwitch</b> parameter | <i>Massive MIMO Uplink Boosting (Low-Frequency TDD)</i> | After uplink MU grouping and pairing and interference randomization under weak uplink coverage are enabled, the proportion of UEs under weak coverage for which frequency selective scheduling is performed may decrease. |

- DMRS interference reduction

| Function Name                                             | Function Switch                                   | Reference                    | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|-----------------------------------------------------------|---------------------------------------------------|------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Frequency-domain multiplexing between PUSCH data and DMRS | <b>NRDUCellPusch.<br/>PuschDmrsFdm<br/>Switch</b> | <i>Uplink<br/>Scheduling</i> | <p>If both DMRS interference reduction and frequency-domain multiplexing between PUSCH data and DMRS are enabled:</p> <ul style="list-style-type: none"><li>• When the proportion of cell-center UEs and the uplink PRB usage are high, the uplink throughput of such UEs may decrease. In this scenario, the DMRS interference reduction function is not recommended .</li><li>• In other scenarios, DMRS interference reduction can be enabled after frequency-domain multiplexing between PUSCH data and DMRS is enabled, but the padding proportion will decrease and accordingly the <b>Cell Uplink</b></li></ul> |

| Function Name | Function Switch | Reference | Description                                   |
|---------------|-----------------|-----------|-----------------------------------------------|
|               |                 |           | <b>Average Throughput (DU)</b> will decrease. |

- High-reliability uplink MCS selection

| Function Name                                                             | Function Switch                                                        | Reference                 | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|---------------------------------------------------------------------------|------------------------------------------------------------------------|---------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Adaptation between UL 256QAM MCS index table and UL 64QAM MCS index table | <b>NRDUCellAlgoSwitch.Ul256QamSwitch</b> set to <b>UL_256QAM_ADAPT</b> | <i>Modulation Schemes</i> | If cell-edge UEs use the UL 256QAM MCS index table, RB estimation is inaccurate. After high-reliability uplink MCS selection is enabled, a possible increase in the uplink QPSK proportion will affect the uplink user-perceived rate of cell-edge UEs. Therefore, it is recommended that adaptation between UL 256QAM MCS index table and UL 64QAM MCS index table be enabled. (Uplink QPSK proportion = $\frac{N_{UL.SCH.QPSK.TB}}{(N_{UL.SCH.QPSK.TB} + N_{UL.SCH.16QAM.TB} + N_{UL.SCH.64QAM.TB} + N_{UL.SCH.256QAM.TB})} \times 100\%$ ) |

| Function Name                                     | Function Switch                                                                          | Reference  | Description                                                                                                                                     |
|---------------------------------------------------|------------------------------------------------------------------------------------------|------------|-------------------------------------------------------------------------------------------------------------------------------------------------|
| Uplink 4-layer transmission for a single FWA user | <b>UL_SU_FOUR_LAYER_SW</b> option of the <b>NRDUCellUIRank.UlRankAlgorithm</b> parameter | <i>FWA</i> | If uplink 4-layer transmission for a single FWA user has taken effect, high-reliability uplink MCS selection does not take effect for the user. |

- Uplink data compression

| Function Name        | Function Switch                                                            | Reference   | Description                                                                                                                                                                                                                                                                                                                                                                      |
|----------------------|----------------------------------------------------------------------------|-------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Basic VoNR functions | <b>VONR_SW</b> option of the <b>NRCCellAlgoSwitch.VonrSwitch</b> parameter | <i>VoNR</i> | If uplink data compression (controlled by the <b>UPLINK_SIG_COMPRESS_SW</b> option of the <b>NRDUCellChnCoverageAlgo.UlCoverageAlgoSwitch</b> parameter or the <b>QCI_UDC_SW</b> option of the <b>gNBQciBearer.PdcpAlgoSwitch</b> parameter) and basic VoNR functions are enabled, voice quality may deteriorate (in terms of extra delay or jitter) or service drops may occur. |

| Function Name | Function Switch                                                                     | Reference   | Description                                                                                                                                                                                                                                                                                                                                                                       |
|---------------|-------------------------------------------------------------------------------------|-------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| MCX Switch    | <b>MCX_SW</b> option of the <b>NRCellAlgoSwitch.ServiceFunctionSwitch</b> parameter | None        | If uplink data compression (controlled by the <b>UPLINK_SIG_COMPRESS_SW</b> option of the <b>NRDUCellChnCoVAlgo.UlCoverageAlgoSwitch</b> parameter or the <b>QCI_UDC_SW</b> option of the <b>gNBQciBearer.PdcpAlgoSwitch</b> parameter) is enabled and the MCX switch is turned on, voice quality may deteriorate (in terms of extra delay or jitter) or service drops may occur. |
| ROHC          | <b>ROHC_SW</b> option of the <b>NRCellAlgoSwitch.RohcSwitch</b> parameter           | <i>VoNR</i> | ROHC and uplink data compression cannot take effect simultaneously for the same bearer of a UE. ROHC preferentially takes effect for the bearer if it is enabled together with uplink data compression.                                                                                                                                                                           |

| Function Name                    | Function Switch                                                                                                       | Reference                          | Description                                                                                                                                                                                                                                                                               |
|----------------------------------|-----------------------------------------------------------------------------------------------------------------------|------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Low latency and high reliability | <b>HIGH_RELIABILITY_BASIC_SWITCH</b> option of the <b>NRDUCellAlgoSwitch</b> . <i>HighReliabilitySwitch</i> parameter | <i>URLLC</i>                       | The low latency and high reliability function and uplink data compression cannot take effect simultaneously for the same bearer of a UE. The low latency and high reliability function preferentially takes effect for the bearer if it is enabled together with uplink data compression. |
| Cloud X services                 | <b>CLOUD_VR_SERVICE_SWITCH</b> option of the <b>NRCeCellAlgoSwitch</b> . <i>CloudVrSwitch</i> parameter               | <i>Ultimate Cloud X Experience</i> | Cloud X services and uplink data compression cannot take effect simultaneously for the same bearer of a UE. Cloud X services preferentially take effect for the bearer if they are enabled together with uplink data compression.                                                         |

- Probability-based uplink MCS selection

| Function Name                    | Function Switch                                                                                                   | Reference         | Description                                                                                                                                                                                                                                                                                                                    |
|----------------------------------|-------------------------------------------------------------------------------------------------------------------|-------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| PUSCH MU-MIMO                    | <b>UL_MU_MIMO_SW</b> option of the <b>NRDUCellAlgoSwitch</b> . <b>MuMimoSwitch</b> parameter                      | <i>MIMO (TDD)</i> | If both PUSCH MU-MIMO and probability-based uplink MCS selection are enabled, cell-edge UEs may be paired, affecting their uplink user-perceived rates. Therefore, it is recommended that the <b>NRDUCellUlimo.UlMuMimoInrThld</b> parameter be set to <b>-20</b> to restrict the pairing of cell-edge UEs for uplink MU-MIMO. |
| Basic VoNR functions             | <b>VONR_SW</b> option of the <b>NRCCellAlgoSwitch</b> . <b>VonrSwitch</b> parameter                               | <i>VoNR</i>       | Probability-based uplink MCS selection does not take effect for UEs with VoNR in effect.                                                                                                                                                                                                                                       |
| Low latency and high reliability | <b>HIGH_RELIABILITY_BASIC_SW</b> option of the <b>NRDUCellAlgoSwitch</b> . <b>HighReliabilitySwitch</b> parameter | <i>URLLC</i>      | For UEs with the low latency and high reliability function in effect, probability-based uplink MCS selection does not take effect.                                                                                                                                                                                             |



| Function Name                   | Function Switch                                                                                     | Reference                                               | Description                                                                                                                                                                  |
|---------------------------------|-----------------------------------------------------------------------------------------------------|---------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Uplink PAMC                     | <b>UL_PAMC_SWITCH</b> option of the <b>NRDUCellPusch.UlPuschAlgoSwitch</b> parameter                | <i>Uplink Scheduling</i>                                | If both probability-based uplink MCS selection and uplink PAMC are enabled, probability-based uplink MCS selection preferentially takes effect in a cell.                    |
| Uplink target IBLER adaptation  | <b>UL_IBLER_ADAPT_SW</b> option of the <b>NRDUCellAlgoSwitch.AdaptiveEdgeExpEnhSwitch</b> parameter | <i>Uplink Scheduling</i>                                | If both probability-based uplink MCS selection and uplink target IBLER adaptation are enabled, probability-based uplink MCS selection preferentially takes effect in a cell. |
| Uplink precise MCS optimization | <b>UL_PRECISE_MCS_OPT_SW</b> option of the <b>NRDUCellAiAlgo.AiAmcAlgoSwitch</b> parameter          | <i>Massive MIMO Uplink Boosting (Low-Frequency TDD)</i> | Uplink precise MCS optimization preferentially takes effect when it is enabled together with probability-based uplink MCS selection.                                         |

- Time-domain adaptive filtering

| Function Name                                         | Function Switch                                                                                                                                                                                                                                              | Reference                                                 | Description                                                                                                                       |
|-------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------|
| Remote interference resistance for PUSCH demodulation | <b>RIM_PUSCH_DEMOD_SW</b> option of the <b>NRDUCellAlgoSwitch.RimAlgoSwitch</b> parameter                                                                                                                                                                    | <i>Remote Interference Management (Low-Frequency TDD)</i> | Time-domain adaptive filtering does not take effect for UEs with remote interference resistance for PUSCH demodulation in effect. |
| UL skipping                                           | <b>UPLINK_SKIP_SW</b> option of the <b>NRDUCellUePwrSaving.SchPwrSavingSw</b> parameter                                                                                                                                                                      | <i>UE Power Saving</i>                                    | Time-domain adaptive filtering does not take effect for UEs with UL skipping in effect.                                           |
| RedCap UL skipping                                    | <ul style="list-style-type: none"> <li><b>REDCAP_UPLINK_SKIP_SW</b> option of the <b>NRDUCellUePwrSaving.SchPwrSavingSw</b> parameter</li> <li><b>REDCAP_UPLINK_SKIP_EXT_SW</b> option of the <b>NRDUCellUePwrSaving.SchPwrSavingSw</b> parameter</li> </ul> | <i>RedCap</i>                                             | Time-domain adaptive filtering does not take effect for UEs with RedCap UL skipping in effect.                                    |

### 5.3.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

#### Base Station Models

3900 and 5900 series base stations (macro base stations). 3900 series base stations must be configured with the BBU3910.

#### Boards

All NR-capable main control boards and NR TDD-capable baseband processing units support this function. For details, see the BBU technical specifications in *3900 & 5900 Series Base Station Product Documentation*.

## RF Modules

Interference randomization under weak uplink coverage/DMRS interference reduction/Uplink data compression:

All NR TDD-capable RF modules that work in low frequency bands support this function. For details, see the technical specifications of RF modules in *3900 & 5900 Series Base Station Product Documentation*.

High-reliability uplink MCS selection/Probability-based uplink MCS selection/Time-domain adaptive filtering:

All NR-capable RF modules with at least 32T32R support this function. For details, see the technical specifications of RF modules in *3900 & 5900 Series Base Station Product Documentation*.

### 5.3.4 Cells

High-reliability uplink MCS selection/Probability-based uplink MCS selection/Time-domain adaptive filtering: Cells must work in 32T32R or 64T64R mode.

Other functions: none

### 5.3.5 Others

Uplink data compression: This function requires that UEs support UDC, which is indicated by the **udc-r17** field.

Uplink data compression based on the SIP standard dictionary: This function requires that UEs support the UDC standard dictionary, which is indicated by the **standardDictionary-r17** field in **udc-r17**.

Other functions: none

## 5.4 Operation and Maintenance

### 5.4.1 Data Configuration

#### 5.4.1.1 Data Preparation

[Table 5-1](#), [Table 5-2](#), [Table 5-3](#), [Table 5-4](#), [Table 5-5](#), [Table 5-6](#), [Table 5-7](#), and [Table 5-8](#) describe the parameters used for function activation. This section does not describe parameters related to cell establishment.

**Table 5-1** Parameters that are recommended to be configured before activating Turbo Uplink 2.0

| Parameter Name                | Parameter ID                                | Option                         | Setting Notes                                                                                                             |
|-------------------------------|---------------------------------------------|--------------------------------|---------------------------------------------------------------------------------------------------------------------------|
| Uplink PUSCH Algorithm Switch | <b>NRDUCellPusch.UlPuschAlgoSwitch</b>      | UL_SCHEDULE_ENH_SW             | To maximize the gains of Turbo Uplink 2.0, it is recommended that this option be selected before this feature is enabled. |
| Uplink PUSCH Algorithm Switch | <b>NRDUCellPusch.UlPuschAlgoSwitch</b>      | UL_INTRF_RANDOMIZED_SCH_ENH_SW | To maximize the gains of Turbo Uplink 2.0, it is recommended that this option be selected before this feature is enabled. |
| UL MU-MIMO Algorithm Switch   | <b>NRDUCellULMimo.ULMuMimoAlgoSwitch</b>    | OPT_PAIR_RB_SEL_SW             | To maximize the gains of Turbo Uplink 2.0, it is recommended that this option be selected before this feature is enabled. |
| PUSCH Performance Switch      | <b>NRDUCellPusch.PuschPerformanceSwitch</b> | UL_FREQ_SEL_SCH_OPT_SW         | To maximize the gains of Turbo Uplink 2.0, it is recommended that this option be selected before this feature is enabled. |
| UL MU-MIMO Algorithm Switch   | <b>NRDUCellULMimo.ULMuMimoAlgoSwitch</b>    | BIG_PKT_ALIGN_PAIR_OPT_SW      | To maximize the gains of Turbo Uplink 2.0, it is recommended that this option be selected before this feature is enabled. |

| Parameter Name                | Parameter ID                                       | Option                    | Setting Notes                                                                                                             |
|-------------------------------|----------------------------------------------------|---------------------------|---------------------------------------------------------------------------------------------------------------------------|
| UL MU-MIMO Algorithm Switch   | <b>NRDUCellULMimo.<i>ULMuMimoAlgoSwitch</i></b>    | SML_PKT_ALIGN_PAIR_OPT_SW | To maximize the gains of Turbo Uplink 2.0, it is recommended that this option be selected before this feature is enabled. |
| Uplink AMC Algorithm Switch   | <b>NRDUCellULAMc.<i>ULAMcAlgoSw</i></b>            | MU_AMC_OPT_SW             | To maximize the gains of Turbo Uplink 2.0, it is recommended that this option be selected before this feature is enabled. |
| Uplink AMC Algorithm Switch   | <b>NRDUCellULAMc.<i>ULAMcAlgoSw</i></b>            | UL_SCH_AVG_SINR_OPT_SW    | To maximize the gains of Turbo Uplink 2.0, it is recommended that this option be selected before this feature is enabled. |
| Uplink PUSCH Algorithm Switch | <b>NRDUCellPusch.<i>ULPuschAlgoSwitch</i></b>      | SR_SMART_SCH_SW           | To maximize the gains of Turbo Uplink 2.0, it is recommended that this option be selected before this feature is enabled. |
| PUSCH Performance Switch      | <b>NRDUCellPusch.<i>PuschPerformanceSwitch</i></b> | UL_NONCONSEC_SCH_OPT_SW   | To maximize the gains of Turbo Uplink 2.0, it is recommended that this option be selected before this feature is enabled. |

| Parameter Name                     | Parameter ID                                | Option                      | Setting Notes                                                                                                                                                                      |
|------------------------------------|---------------------------------------------|-----------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| MU RB Ratio Threshold              | <b>NRDUCellUIMimo.MuRbRatioThld</b>         | None                        | Set this parameter based on the network plan. To maximize the gains of Turbo Uplink 2.0, it is recommended that this parameter be set to <b>40</b> before this feature is enabled. |
| UL MU-MIMO SINR Threshold          | <b>NRDUCellUIMimo.UIMuMimoSinrThld</b>      | None                        | To maximize the gains of Turbo Uplink 2.0, it is recommended that this parameter be set to <b>-20</b> before this feature is enabled.                                              |
| PDCCH Algorithm Enhancement Switch | <b>NRDUCellPdcch.PdcchAlgoEnhSwitch</b>     | PUSCH_DTX_AGG_LVL_ADAPT_SW  | To maximize the gains of Turbo Uplink 2.0, it is recommended that this option be deselected before this feature is enabled.                                                        |
| PUSCH Performance Switch           | <b>NRDUCellPusch.PuschPerformanceSwitch</b> | UL_RB_ESTIMATE_OPT_SW       | To maximize the gains of Turbo Uplink 2.0, it is recommended that this option be selected before this feature is enabled.                                                          |
| Uplink AMC Performance Switch      | <b>NRDUCellUIAmc.UIAmcPerformanceSw</b>     | DUAL_FILTER_FREQ_SEL_OPT_SW | To maximize the gains of Turbo Uplink 2.0, it is recommended that this option be selected before this feature is enabled.                                                          |

| Parameter Name                | Parameter ID                             | Option                       | Setting Notes                                                                                                                                     |
|-------------------------------|------------------------------------------|------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------|
| Uplink 256QAM Switch          | <b>NRDUCellAlgoSwitch.Ul256QamSwitch</b> | None                         | To maximize the gains of Turbo Uplink 2.0, it is recommended that this parameter be set to <b>UL_256QAM_ADAPT</b> before this feature is enabled. |
| Uplink AMC Algorithm Switch   | <b>NRDUCellUlamc.UlAmcAlgoSw</b>         | UL_256QAM_ADAPT_POWER_OPT_SW | To maximize the gains of Turbo Uplink 2.0, it is recommended that this option be selected before this feature is enabled.                         |
| Uplink AMC Performance Switch | <b>NRDUCellUlamc.UlAmcPerformanceSw</b>  | LOST_UE_DROP_PROTECT_SW      | To maximize the gains of Turbo Uplink 2.0, it is recommended that this option be selected before this feature is enabled.                         |

**Table 5-2** Parameters used for activating Turbo Uplink 2.0

| Parameter Name               | Parameter ID                                        | Option                 | Setting Notes                                                   |
|------------------------------|-----------------------------------------------------|------------------------|-----------------------------------------------------------------|
| UL Coverage Algorithm Switch | <b>NRDUCellChnCovAlgo.UlCoverageAlgorithmSwitch</b> | TURBO_UPLINK_PHASE2_SW | Select this option if the Turbo Uplink 2.0 feature is required. |

**Table 5-3** Parameters used for activating interference randomization under weak uplink coverage

| Parameter Name               | Parameter ID                                   | Option                        | Setting Notes                                                                                                                     |
|------------------------------|------------------------------------------------|-------------------------------|-----------------------------------------------------------------------------------------------------------------------------------|
| UL Coverage Algorithm Switch | <b>NRDUCellChnCovAlgo.UlCoverageAlgoSwitch</b> | UL_WEAK_COV_I<br>NTRF_RAND_SW | Select this option if interference randomization under weak uplink coverage is required.                                          |
| SRS RSRP Exception Threshold | <b>NRDUCellSrsMeas.SrsRsrpExcThld</b>          | None                          | It is recommended that this parameter be set to <b>DB6</b> when interference randomization under weak uplink coverage is enabled. |
| UL Coverage Algorithm Switch | <b>NRDUCellChnCovAlgo.UlCoverageAlgoSwitch</b> | UL_RETRANS_ALL<br>OC_ADJ_SW   | Select this option for cells where interference randomization under weak uplink coverage needs to be enabled.                     |

**Table 5-4** Parameters used for activating DMRS interference reduction

| Parameter Name               | Parameter ID                                   | Option                   | Setting Notes       |
|------------------------------|------------------------------------------------|--------------------------|---------------------|
| UL Coverage Algorithm Switch | <b>NRDUCellChnCovAlgo.UlCoverageAlgoSwitch</b> | DMRS_RESOURCE<br>_ADJ_SW | Select this option. |



**Table 5-5** Parameters used for activating high-reliability uplink MCS selection

| Parameter Name                | Parameter ID                       | Option                        | Setting Notes                                                                         |
|-------------------------------|------------------------------------|-------------------------------|---------------------------------------------------------------------------------------|
| Uplink AMC Performance Switch | <b>NRDUCellUplinkPerformanceSw</b> | UL_HIGH_RELIAB_MCS_SEL_SW     | Select this option for cells where high-reliability uplink MCS selection is required. |
| Uplink AMC Performance Switch | <b>NRDUCellUplinkPerformanceSw</b> | UL_DMRS_FILTER_COEFF_ADAPT_SW | Select this option for cells where high-reliability uplink MCS selection is required. |

**Table 5-6** Parameters used for activating uplink data compression

| Parameter Name               | Parameter ID                              | Option                 | Setting Notes                                                                                                                   |
|------------------------------|-------------------------------------------|------------------------|---------------------------------------------------------------------------------------------------------------------------------|
| UL Coverage Algorithm Switch | <b>NRDUCellChnCoverageAlgorithmSwitch</b> | UDC_SW                 | Select this option if uplink data compression is required for all uplink data packets transmitted in a cell for data services.  |
| UL Coverage Algorithm Switch | <b>NRDUCellChnCoverageAlgorithmSwitch</b> | UPLINK_SIG_COMPRESS_SW | Select this option if uplink data compression is required for all uplink data packets transmitted in a cell for voice services. |

| Parameter Name         | Parameter ID                               | Option             | Setting Notes                                                                                                                                                                                                                                                                                                                                                              |
|------------------------|--------------------------------------------|--------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| PDCP Algorithm Switch  | <b><i>gNBQciBearer.PdcpAlgoSwitch</i></b>  | QCI_UDC_SW         | If uplink data compression is required only for uplink data packets carried on bearers with the specified QCI, deselect the <b>UDC_SW</b> and <b>UPLINK_SIG_COMPRESS_SW</b> options of the <b>NRDUCellChnCovAlgo.UlCoverageAlgoSwitch</b> parameter and select the <b>QCI_UDC_SW</b> option of the <b>gNBQciBearer.PdcpAlgoSwitch</b> parameter for the corresponding QCI. |
| UDC Buffer Size        | <b><i>NRCelServExp.UdcBufferSize</i></b>   | None               | The default value is recommended when uplink data compression is required.                                                                                                                                                                                                                                                                                                 |
| Voice Algorithm Switch | <b><i>NRCelServExp.VoiceAlgoSwitch</i></b> | UL_SIP_UDC_DICT_SW | Select this option if uplink data compression based on the SIP standard dictionary is required.                                                                                                                                                                                                                                                                            |

**Table 5-7** Parameters used for activating probability-based uplink MCS selection

| Parameter Name                          | Parameter ID                               | Option                        | Setting Notes                                                                          |
|-----------------------------------------|--------------------------------------------|-------------------------------|----------------------------------------------------------------------------------------|
| Uplink AMC Performance Switch           | <b>NRDUCellUIMc.UlAmcPerformanceSw</b>     | UL_PBMCSSEL_SW                | Select this option for cells where probability-based uplink MCS selection is required. |
| Uplink AMC Performance Switch           | <b>NRDUCellUIMc.UlAmcPerformanceSw</b>     | UL_HIGH_RELIAB_MCS_SEL_SW     | Select this option for cells where probability-based uplink MCS selection is required. |
| UL Scheduling Resource Algorithm Switch | <b>NRDUCellUISchRes.UlSchResAlgoSwitch</b> | MCS_TABLE_BASED_SCH_RB_ADJ_SW | Select this option for cells where probability-based uplink MCS selection is required. |

**Table 5-8** Parameters used for activating time-domain adaptive filtering

| Parameter Name               | Parameter ID                                   | Option                            | Setting Notes                                                                  |
|------------------------------|------------------------------------------------|-----------------------------------|--------------------------------------------------------------------------------|
| UL Coverage Algorithm Switch | <b>NRDUCellChnCovAlgo.UlCoverageAlgoSwitch</b> | PUSCH_TIME_DOMAIN_ADAPT_FILTER_SW | Select this option for cells where time-domain adaptive filtering is required. |

### 5.4.1.2 Using MML Commands

Before using MML commands, refer to [5.3 Requirements](#) and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

## Activation Command Examples

- To maximize the gains provided by this feature, it is recommended that the following operations be performed before this feature is enabled:

```
//Enabling uplink scheduling enhancement
MOD NRDUCELLPUSCH: NrDuCellId=0, UIPuschAlgoSwitch=UL_SCHEDULE_ENH_SW-1;
//Enabling uplink interference randomization-based scheduling enhancement
MOD NRDUCELLPUSCH: NrDuCellId=0,
UIPuschAlgoSwitch=UL_INTRF_RANDOMIZED_SCH_ENH_SW-1;
//Enabling optimal paired RB selection
MOD NRDUCELLULMIMO: NrDuCellId=0, UIMuMimoAlgoSwitch=OPT_PAIR_RB_SEL_SW-1;
//Enabling uplink frequency selective scheduling optimization
MOD NRDUCELLPUSCH: NrDuCellId=0, PuschPerformanceSwitch=UL_FREQ_SEL_SCH_OPT_SW-1;
//Enabling alignment pairing optimization for large-packet UEs
MOD NRDUCELLULMIMO: NrDuCellId=0, UIMuMimoAlgoSwitch=BIG_PKT_ALIGN_PAIR_OPT_SW-1;
//Enabling alignment pairing optimization for small-packet UEs
MOD NRDUCELLULMIMO: NrDuCellId=0, UIMuMimoAlgoSwitch=SML_PKT_ALIGN_PAIR_OPT_SW-1;
//Selecting the MU_AMC_OPT_SW option
MOD NRDUCELLULAMC: NrDuCellId=0, UIAmcAlgoSw=MU_AMC_OPT_SW-1;
//Enabling average SINR optimization in uplink scheduling
MOD NRDUCELLULAMC: NrDuCellId=0, UIAmcAlgoSw=UL_SCH_AVG_SINR_OPT_SW-1;
//Enabling SR-based smart scheduling
MOD NRDUCELLPUSCH: NrDuCellId=0, UIPuschAlgoSwitch=SR_SMART_SCH_SW-1;
//Enabling uplink non-consecutive scheduling optimization
MOD NRDUCELLPUSCH: NrDuCellId=0, PuschPerformanceSwitch=UL_NONCONSEC_SCH_OPT_SW-1;
//Setting the MuRbRatioThld parameter (not required if this parameter has been set to 60 on the live network)
MOD NRDUCELLULMIMO: NrDuCellId=0, MuRbRatioThld=40;
//Setting the UIMuMimoSinrThld parameter
MOD NRDUCELLULMIMO: NrDuCellId=0, UIMuMimoSinrThld=-20;
//Disabling PUSCH-DTX-based aggregation level adaptation
MOD NRDUCELLPDCCH: NrDuCellId=0, PdcchAlgoEnhSwitch=PUSCH_DTX_AGG_LVL_ADAPT_SW-0;
//Enabling uplink RB estimation optimization
MOD NRDUCELLPUSCH: NrDuCellId=0, PuschPerformanceSwitch=UL_RB_ESTIMATE_OPT_SW-1;
//Enabling frequency selective scheduling optimization in dual filtering scenarios
MOD NRDUCELLULAMC: NrDuCellId=0, UIAmcPerformanceSw=DUAL_FILTER_FREQ_SEL_OPT_SW-1;
//Setting the UL256QamSwitch parameter
MOD NRDUCELLALGOSWITCH: NrDuCellId=0, UI256QamSwitch=UL_256QAM_ADAPT;
//Selecting the UL_256QAM_ADAPT_POWER_OPT_SW option
MOD NRDUCELLULAMC: NrDuCellId=0, UIAmcAlgoSw=UL_256QAM_ADAPT_POWER_OPT_SW-1;
//Enabling lost-UE protection against service drops
MOD NRDUCELLULAMC: NrDuCellId=0, UIAmcPerformanceSw=LOST_UE_DROP_PROTECT_SW-1;
```

- Turbo Uplink 2.0

```
//Enabling the Turbo Uplink 2.0 feature
MOD NRDUCELLCHNCOVALGO: NrDuCellId=0,
UICoverageAlgoSwitch=TURBO_UPLINK_PHASE2_SW-1;
```

- Interference randomization under weak uplink coverage

```
//Enabling interference randomization under weak uplink coverage
MOD NRDUCELLCHNCOVALGO: NrDuCellId=0,
UICoverageAlgoSwitch=UL_WEAK_COV_INTRF RAND_SW-1;
//Setting the SRS RSRP Exception Threshold parameter
MOD NRDUCELLSRSMEAS: NrDuCellId=0, SrsRsrpExcThld=DB6;
//Enabling allocation adjustment for uplink retransmission
MOD NRDUCELLCHNCOVALGO: NrDuCellId=0,
UICoverageAlgoSwitch=UL_RETRANS_ALLOC_ADJ_SW-1;
```

- DMRS interference reduction

```
//Enabling DMRS interference reduction
MOD NRDUCELLCHNCOVALGO: NrDuCellId=0, UICoverageAlgoSwitch=DMRS_RESOURCE_ADJ_SW-1;
```

- High-reliability uplink MCS selection

```
//Selecting the UL_HIGH_RELIAB_MCS_SEL_SW option
MOD NRDUCELLULAMC: NrDuCellId=0, UIAmcPerformanceSw=UL_HIGH_RELIAB_MCS_SEL_SW-1;
//Selecting the UL_DMRS_FILTER_COEFF_ADAPT_SW option
MOD NRDUCELLULAMC: NrDuCellId=0,
UIAmcPerformanceSw=UL_DMRS_FILTER_COEFF_ADAPT_SW-1;
```

- Uplink data compression

- **Cell-level uplink data compression**  
//Selecting the UDC\_SW option if uplink data compression is required for all uplink data packets transmitted in a cell for data services  
MOD NRDUCELLCHNCOVALGO: NrDuCellId=0, UICoverageAlgoSwitch=UDC\_SW-1;  
//Selecting the UPLINK\_SIG\_COMPRESS\_SW option if uplink data compression is required for all uplink data packets transmitted in a cell for voice services  
MOD NRDUCELLCHNCOVALGO: NrDuCellId=0, UICoverageAlgoSwitch=UPLINK\_SIG\_COMPRESS\_SW-1;
- **QCI-specific uplink data compression**  
//Deselecting the UDC\_SW and UPLINK\_SIG\_COMPRESS\_SW options and selecting the QCI\_UDC\_SW option if uplink data compression is required only for uplink data packets carried on bearers with the specified QCI (for example, QCI 8)  
MOD NRDUCELLCHNCOVALGO: NrDuCellId=0, UICoverageAlgoSwitch=UDC\_SW-0;  
MOD NRDUCELLCHNCOVALGO: NrDuCellId=0, UICoverageAlgoSwitch=UPLINK\_SIG\_COMPRESS\_SW-0;  
MOD GNBQCIBEARER: Qci=8, PdcAlgoSwitch=QCI\_UDC\_SW-1;
- **Uplink data compression based on the SIP standard dictionary**  
//Selecting the UL\_SIP\_UDC\_DICT\_SW option  
MOD NRCELLSERVEXP: NrCellId=0, VoiceAlgoSwitch=UL\_SIP\_UDC\_DICT\_SW-1;
- **Other parameters**  
//Configuring the buffer size for uplink data compression  
MOD NRCELLSERVEXP: NrCellId=0, UdcBufferSize=8KBYTE;
- **Probability-based uplink MCS selection**  
//Enabling probability-based uplink MCS selection  
MOD NRDUCELLULAMC: NrDuCellId=0, UIAmcPerformanceSw=UL\_PBMCSSEL\_SW-1;  
//Selecting the UL\_HIGH\_RELIAB\_MCS\_SEL\_SW option  
MOD NRDUCELLULAMC: NrDuCellId=0, UIAmcPerformanceSw=UL\_HIGH\_RELIAB\_MCS\_SEL\_SW-1;  
//Selecting the MCS\_TABLE\_BASED\_SCH\_RB\_ADJ\_SW option  
MOD NRDUCELLULSCHRES: NrDuCellId=0, UISchResAlgoSwitch=MCS\_TABLE\_BASED\_SCH\_RB\_ADJ\_SW-1;
- **Time-domain adaptive filtering**  
//Enabling time-domain adaptive filtering  
MOD NRDUCELLCHNCOVALGO: NrDuCellId=0, UICoverageAlgoSwitch=PUSCH\_TIME\_DOM\_ADAPT\_FILTER\_SW-1;

## Deactivation Command Examples

The following provides only deactivation command examples. You can determine whether to restore the settings of other parameters based on actual network conditions.

- **Interference randomization under weak uplink coverage**  
//Disabling allocation adjustment for uplink retransmission  
MOD NRDUCELLCHNCOVALGO: NrDuCellId=0, UICoverageAlgoSwitch=UL\_RETRANS\_ALLOC\_ADJ\_SW-0;  
//Disabling interference randomization under weak uplink coverage  
MOD NRDUCELLCHNCOVALGO: NrDuCellId=0, UICoverageAlgoSwitch=UL\_WEAK\_COV\_INTRF\_RAND\_SW-0;
- **DMRS interference reduction**  
//Disabling DMRS interference reduction  
MOD NRDUCELLCHNCOVALGO: NrDuCellId=0, UICoverageAlgoSwitch=DMRS\_RESOURCE\_ADJ\_SW-0;
- **High-reliability uplink MCS selection**  
//Deselecting the UL\_DMRS\_FILTER\_COEFF\_ADAPT\_SW option  
MOD NRDUCELLULAMC: NrDuCellId=0, UIAmcPerformanceSw=UL\_DMRS\_FILTER\_COEFF\_ADAPT\_SW-0;  
//Deselecting the UL\_HIGH\_RELIAB\_MCS\_SEL\_SW option  
MOD NRDUCELLULAMC: NrDuCellId=0, UIAmcPerformanceSw=UL\_HIGH\_RELIAB\_MCS\_SEL\_SW-0;
- **Uplink data compression**
  - **Uplink data compression based on the SIP standard dictionary**  
//Deselecting the UL\_SIP\_UDC\_DICT\_SW option  
MOD NRCELLSERVEXP: NrCellId=0, VoiceAlgoSwitch=UL\_SIP\_UDC\_DICT\_SW-0;

- Cell-level uplink data compression  
//Deselecting the UDC\_SW option  
MOD NRDUCELLCHNCOVALGO: NrDuCellId=0, UICoverageAlgoSwitch=UDC\_SW-0;  
//Deselecting the UPLINK\_SIG\_COMPRESS\_SW option  
MOD NRDUCELLCHNCOVALGO: NrDuCellId=0,  
UICoverageAlgoSwitch=UPLINK\_SIG\_COMPRESS\_SW-0;
- QCI-specific uplink data compression  
//Deselecting the QCI\_UDC\_SW option for the corresponding QCI  
MOD GNBQCIBEARER: Qci=8, PdcchAlgoSwitch=QCI\_UDC\_SW-0;
- Probability-based uplink MCS selection  
//Disabling probability-based uplink MCS selection  
MOD NRDUCELLULAMC: NrDuCellId=0, UIAmcPerformanceSw=UL\_PBMCSSEL\_SW-0;  
//Deselecting the UL\_HIGH\_RELIAB\_MCS\_SEL\_SW option  
MOD NRDUCELLULAMC: NrDuCellId=0, UIAmcPerformanceSw=UL\_HIGH\_RELIAB\_MCS\_SEL\_SW-0;  
//Deselecting the MCS\_TABLE\_BASED\_SCH\_RB\_ADJ\_SW option  
MOD NRDUCELLULSCHRES: NrDuCellId=0,  
UISchResAlgoSwitch=MCS\_TABLE\_BASED\_SCH\_RB\_ADJ\_SW-0;
- Time-domain adaptive filtering  
//Disabling time-domain adaptive filtering  
MOD NRDUCELLCHNCOVALGO: NrDuCellId=0,  
UICoverageAlgoSwitch=PUSCH\_TIME\_DOM\_ADAPT\_FILTER\_SW-0;
- Turbo Uplink 2.0  
//Disabling the Turbo Uplink 2.0 feature  
MOD NRDUCELLCHNCOVALGO: NrDuCellId=0,  
UICoverageAlgoSwitch=TURBO\_UPLINK\_PHASE2\_SW-0;
- Determine whether to restore the settings of certain parameters by performing the following operations based on network conditions:  
//Disabling uplink scheduling enhancement  
MOD NRDUCELLPUSCH: NrDuCellId=0, UIPuschAlgoSwitch=UL\_SCHEDULE\_ENH\_SW-0;  
//Disabling uplink interference randomization-based scheduling enhancement  
MOD NRDUCELLPUSCH: NrDuCellId=0,  
UIPuschAlgoSwitch=UL\_INTRF\_RANDOMIZED\_SCH\_ENH\_SW-0;  
//Disabling optimal paired RB selection  
MOD NRDUCELLULMIMO: NrDuCellId=0, UIMuMimoAlgoSwitch=OPT\_PAIR\_RB\_SEL\_SW-0;  
//Disabling uplink frequency selective scheduling optimization  
MOD NRDUCELLPUSCH: NrDuCellId=0, PuschPerformanceSwitch=UL\_FREQ\_SEL\_SCH\_OPT\_SW-0;  
//Disabling alignment pairing optimization for large-packet UEs  
MOD NRDUCELLULMIMO: NrDuCellId=0, UIMuMimoAlgoSwitch=BIG\_PKT\_ALIGN\_PAIR\_OPT\_SW-0;  
//Disabling alignment pairing optimization for small-packet UEs  
MOD NRDUCELLULMIMO: NrDuCellId=0, UIMuMimoAlgoSwitch=SML\_PKT\_ALIGN\_PAIR\_OPT\_SW-0;  
//Deselecting the MU\_AMC\_OPT\_SW option  
MOD NRDUCELLULAMC: NrDuCellId=0, UIAmcAlgoSw=MU\_AMC\_OPT\_SW-0;  
//Disabling average SINR optimization in uplink scheduling  
MOD NRDUCELLULAMC: NrDuCellId=0, UIAmcAlgoSw=UL\_SCH\_AVG\_SINR\_OPT\_SW-0;  
//Disabling SR-based smart scheduling  
MOD NRDUCELLPUSCH: NrDuCellId=0, UIPuschAlgoSwitch=SR\_SMART\_SCH\_SW-0;  
//Disabling uplink non-consecutive scheduling optimization  
MOD NRDUCELLPUSCH: NrDuCellId=0, PuschPerformanceSwitch=UL\_NONCONSEC\_SCH\_OPT\_SW-0;  
//Restoring the MuRbRatioThld parameter  
MOD NRDUCELLULMIMO: NrDuCellId=0, MuRbRatioThld=0;  
//Restoring the UIMuMimoSinrThld parameter  
MOD NRDUCELLULMIMO: NrDuCellId=0, UIMuMimoSinrThld=-200;  
//Enabling PUSCH-DTX-based aggregation level adaptation  
MOD NRDUCELLPDCCH: NrDuCellId=0, PdcchAlgoEnhSwitch=PUSCH\_DTX\_AGG\_LVL\_ADAPT\_SW-1;  
//Disabling uplink RB estimation optimization  
MOD NRDUCELLPUSCH: NrDuCellId=0, PuschPerformanceSwitch=UL\_RB\_ESTIMATE\_OPT\_SW-0;  
//Disabling frequency selective scheduling optimization in dual filtering scenarios  
MOD NRDUCELLULAMC: NrDuCellId=0, UIAmcPerformanceSw=DUAL\_FILTER\_FREQ\_SEL\_OPT\_SW-0;  
//Restoring the UI256QamSwitch parameter  
MOD NRDUCELLALGOSWITCH: NrDuCellId=0, UI256QamSwitch=UL\_256QAM\_OFF;  
//Deselecting the UL\_256QAM\_ADAPT\_POWER\_OPT\_SW option  
MOD NRDUCELLULAMC: NrDuCellId=0, UIAmcAlgoSw=UL\_256QAM\_ADAPT\_POWER\_OPT\_SW-0;  
//Disabling lost-UE protection against service drops  
MOD NRDUCELLULAMC: NrDuCellId=0, UIAmcPerformanceSw=LOST\_UE\_DROP\_PROTECT\_SW-0;

### 5.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

 **NOTE**

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

## 5.4.2 Activation Verification

Uplink data compression:

If the compression efficiency of uplink data packets increases after this function is enabled, this function has taken effect.

Uplink data compression based on the SIP standard dictionary:

- Step 1** On the MAE-Access, choose **Monitor > Signaling Trace > Signaling Trace Management**.
- Step 2** In the navigation tree on the left of the **Signaling Trace Management** tab page, choose **NR > Application Layer > Uu Interface Trace**.
- Step 3** If the value of **dictionary-r17** is **sip-SDP** in the RRCReconfiguration message for setup of bearers of 5QI 5 and 5QI 69, this function has taken effect.

----End

Any other function of Turbo Uplink 2.0:

If the uplink user-perceived rate at the coverage edge increases after this function is enabled, this function has taken effect.

 **NOTE**

Compression efficiency =  $(1 - \text{N.PDCP.Vol.Ul.UDC.BefDecomp} / \text{N.PDCP.Vol.Ul.UDC.AftDecomp}) \times 100\%$

Uplink user-perceived rate at the coverage edge =  $(\text{N.ThpVol.Ul.Edge} - \text{N.ThpVol.Ue.Ul.SmallPkt.Edge}) / \text{N.ThpTime.Ue.Ul.RmvSmallPkt.Edge}$

## 5.4.3 Network Monitoring

The performance indicators listed in [5.2 Network Analysis](#) can be used to monitor the running status of functions of Turbo Uplink 2.0.

# 6 Parameters

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The following hyperlinked EXCEL files of parameter reference match the software version with which this document is released.

- Node Parameter Reference: contains device and transport parameters.
- Node Used Reserved Parameter List: contains the reserved parameters that are in use and those that have been disused.
- gNodeBFunction Parameter Reference: contains all parameters related to radio access functions, including air interface management, access control, mobility control, and radio resource management.
- 3900 & 5900 Series Base Station gNodeBFunction Used Reserved Parameter List: contains the reserved parameters that are in use and the reserved parameters that fall into disuse in the current R version.

## NOTE

You can find the EXCEL files of parameter reference and used reserved parameter list for the software version used on the live network from the "Man-Machine Interface Reference" node in the *3900 & 5900 Series Base Station Product Documentation* delivered with that version.

## **FAQ 1: How do I find the parameters related to a certain feature from parameter reference?**

- Step 1** Open the EXCEL file of parameter reference.
- Step 2** On the **Parameter List** sheet, filter the **Feature ID** column. Click **Text Filters** and choose **Contains**. Enter the feature ID, for example, FBFD-010011.
- Step 3** Click **OK**. All parameters related to the feature are displayed.

----End

## **FAQ 2: How do I find the information about a certain reserved parameter from the used reserved parameter list?**

- Step 1** Open the EXCEL file of the used reserved parameter list.



**Step 2** On the **Used Reserved Parameter List** sheet, use the **MO**, **Parameter ID**, and **BIT** columns to locate the reserved parameter. View its information, including the meaning, values, and impacts.

----End

# 7 Counters

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The following hyperlinked EXCEL files of performance counter reference match the software version with which this document is released.

- Node Performance Counter Summary: contains device and transport counters.
- gNodeBFunction Performance Counter Summary: contains all counters related to radio access functions, including air interface management, access control, mobility control, and radio resource management.
- gNodeBFunction Used Reserved Counter List: contains the reserved counters that are in use and those that have been disused.

## NOTE

You can find the EXCEL files of performance counter reference and used reserved counter list for the software version used on the live network from the "Man-Machine Interface Reference" node in the *3900 & 5900 Series Base Station Product Documentation* delivered with that version.

## **FAQ: How do I find the counters related to a certain feature from performance counter reference?**

- Step 1** Open the EXCEL file of performance counter reference.
- Step 2** On the **Counter Summary(En)** sheet, filter the **Feature ID** column. Click **Text Filters** and choose **Contains**. Enter the feature ID, for example, FBFD-010011.
- Step 3** Click **OK**. All counters related to the feature are displayed.

----End

# 8 Glossary

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For the acronyms, abbreviations, terms, and definitions, see *Glossary*.

# 9 Reference Documents

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- 3GPP TS 38.101-1: "NR; User Equipment (UE) radio transmission and reception"
- 3GPP TS 38.214: "NR; Physical layer procedures for data"
- 3GPP TS 38.104: "NR; Base Station (BS) radio transmission and reception"
- *High Speed Mobility*
- *Hyper Cell*
- *Cell Combination*
- *MIMO (TDD)*
- *Modulation Schemes*
- *Uplink Scheduling*
- *Cell Management*
- *Distributed Antenna Solutions (Low-Frequency TDD)*
- *FWA*
- *3900 & 5900 Series Base Station Product Documentation*
- *Remote Interference Management (Low-Frequency TDD)*
- *URLLC*
- *Channel Management*
- *UE Power Saving*
- *VoNR*
- *License Control Item Lists*
- *Fusion Cell (Low-Frequency TDD)*
- *RedCap*
- *Massive MIMO Uplink Boosting (Low-Frequency TDD)*
- *Ultimate Cloud X Experience*
- *QoS Management*
- *Ultra Uplink (High-Frequency TDD)*